Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Physics Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-II) Physics. It contains comments on candidates' responses to the 2017 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

In questions that required short written answers, most candidates had knowledge of the correct answer, but were undisciplined in communicating that knowledge. This was particularly evident in Question 2, and 7. Candidates who considered how to communicate their knowledge in all types of questions were generally much more successful. Communication includes drawing field lines and vectors, labelling diagrams, using exact terminology, using correct formulae and answering the specific question asked.

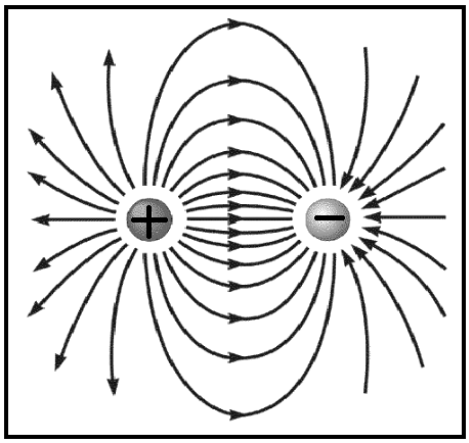
This year candidates performed well on questions related to the electrostatics, electromagnetism, electromagnetic induction, physics of solids, electronics and nuclear physics. Whereas low-scoring candidates did not perform well in questions based on current electricity, dawn of modern physics, and atomic spectra, candidates also had problem in articulating their answers on differences, reasoning and discussion type questions.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

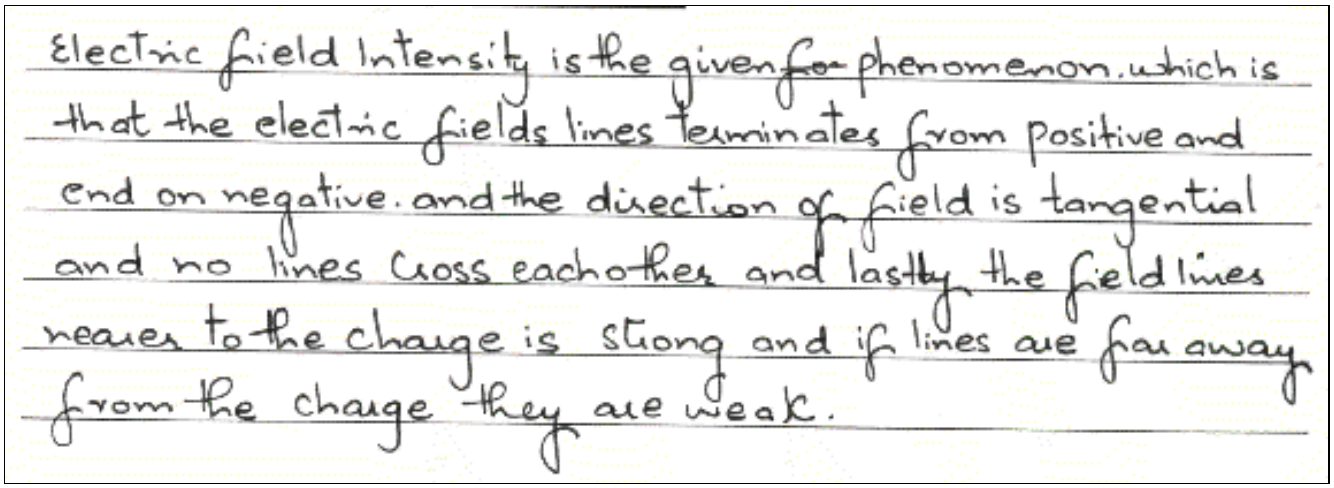
Name and describe the phenomenon illustrated in the given diagram.



Most candidates attempted this question well.

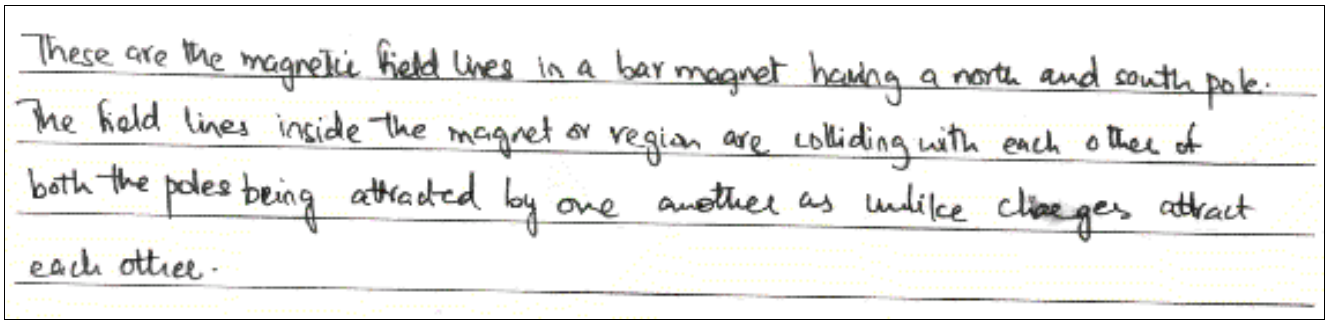
Better responses presented correct name of the phenomenon, i.e. electric dipole, attractive force, electrostatic force and electrostatic induction. High scoring candidates described properly the phenomenon illustrated in the diagram, i.e. electric dipole composes of two electric charges of equal magnitude but opposite sign having a small separation between positive and negative charges; electric field is produced by two equal and opposite charges when two charges are brought closer to each other they experience electrostatic or coulomb's force.

Example:


Electric field Intensity is the given for phenomenon. which is that the electric fields lines terminates from positive and end on negative. and the direction of field is tangential and no lines cross each other and lastly the field lines neares to the charge is stiong and if lines are far away from the charge they are weak.

Weaker responses presented incorrect name of the phenomenon. Candidates identified the phenomenon as electromagnetic induction/ conventional current. Low scoring candidates described bar magnet, poles of a bar magnet and the attraction and repulsion between the two poles of a bar magnet, which were not the demand of the question.

Example:


These are the magnetic field lines in a bar magnet having a north and south pole. The field lines inside the magnet or region are colliding with each other of both the poles being attracted by one another as unlike charges attract each other.

Question 2:

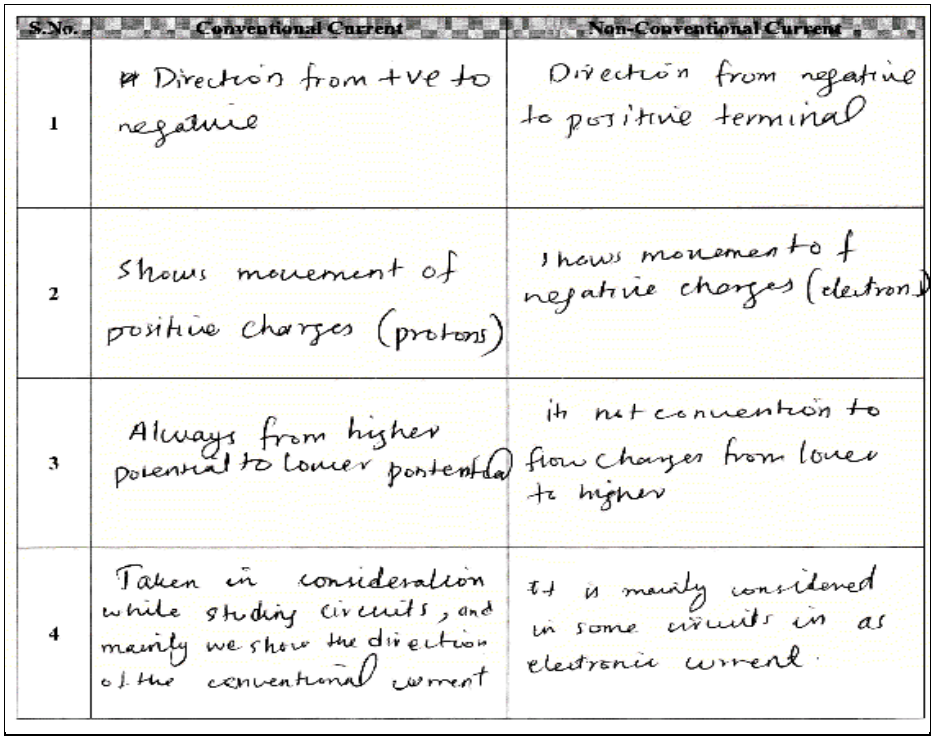
Differentiate between conventional and non-conventional current.

S. No.	Conventional Current	Non-Conventional Current
1		
2		
3		
4		

Most candidates found this question challenging and hence the performance was below expectations.

Better responses differentiated between conventional and non-conventional current in a well-organised manner, some of the candidates drew circuit diagrams to show the direction of flow of conventional and non-conventional current. High scoring candidates wrote differentiation on the basis of valence-bond theory.

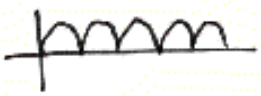
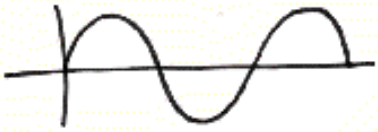
Example:



S.No.	Conventional Current	Non-Conventional Current
1	* Direction from +ve to negative	Direction from negative to positive terminal
2	Shows movement of positive charges (protons)	Shows movement of negative charges (electrons)
3	Always from higher potential to lower potential	it is not convention to flow charges from lower to higher
4	Taken in consideration while studying circuits, and mainly we show the direction of the conventional current	It is mainly considered in some circuits in as electronic current.

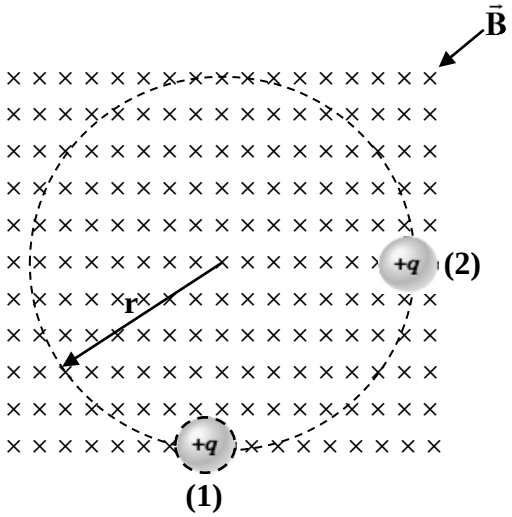
Weaker responses failed to differentiate between conventional and non-conventional current in all four points. Some of the candidates wrote the names of the physicist who discovered electric current. It was observed that in most cases, candidates were unable to write all four valid differences. Some of the low scoring candidates wrote 'direct and alternating current, their direction and polarity' which were irrelevant to the demand of the question.

Example:

S.No.	Conventional Current	Non-Conventional Current
1	Conventional current is also known as direct current.	Non-Conventional current in other words is alter-nating current.
2	It's ~~has~~ ~~a~~ graphical representation is 	Graphical Representation for A.C 
3	The main sources of this are battries and it's ~~mag~~ not used in everyday life	This current is used in everyday life. ~~It~~ Also keeps on reversing it's polarity by the time.
4	This was discovered by Thomas Edison. It's expensive than A.C current.	Created by Nicola Tesla. Cheap as compared to DC current and widely used.

Question 3:

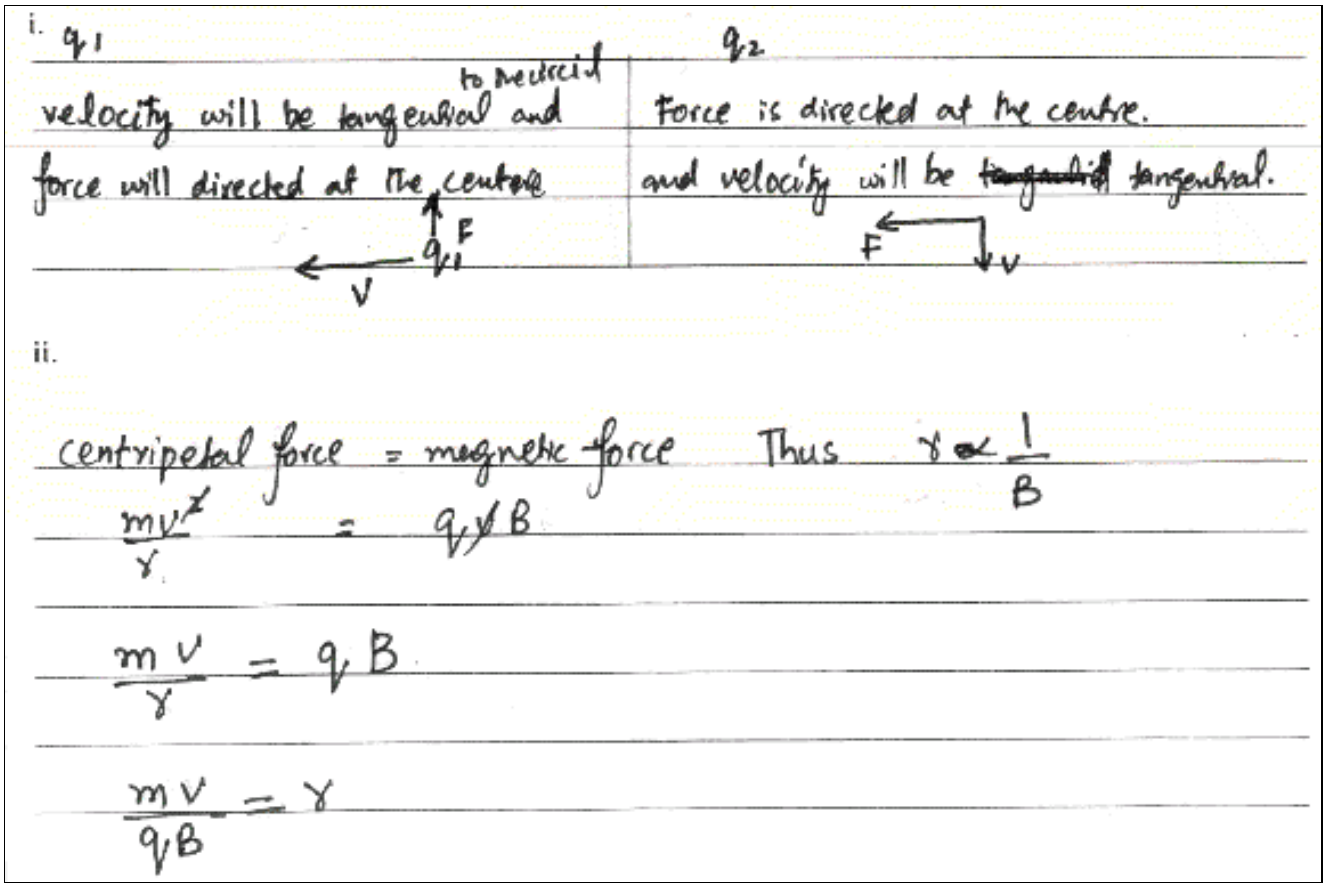
i. Magnetic force helps to move a particle in a circular orbit. If a charged particle ***+q*** is moving in a circular orbit with radius **r** in a magnetic field $\vec{\mathbf{B}}$, then compare the particle's velocity $\vec{\mathbf{v}}$ and the magnetic force $\vec{\mathbf{F}}$ acting on it at point **(1)** and point **(2)**.



ii. Derive a mathematical relation to show that the radius of the given circle is inversely proportional to the magnitude of the magnetic field.

Better responses correctly compared the velocity and magnetic force acting on both points, e.g. when a positively charged particle is at point (1), the magnetic force is perpendicular to the velocity and points directly upward in the diagram and likewise when the particle reaches at point (2), the magnetic force still remains perpendicular to the velocity. In the second part of question, high scoring candidates correctly derived the a mathematical relation to show that the radius of the given circle is inversely proportional to the magnitude of the magnetic field, for this purpose they wrote $F_c = \frac{mv^2}{r}$, $qvB = \frac{mv^2}{r}$, $B = \frac{mv}{qr}$ or $r = \frac{mv}{qB}$.

Example:



Weaker responses indicated that candidates were not able to compare the velocity and magnetic force. Most candidates derived the formula of force on a unit charge in a magnetic field, and at point (1) they wrote that velocity and force will be positive and at point (2), both will be negative. In the second part, most candidates understood the demand of the question and tried to derive the mathematical relation and reached the conclusion expression, $r = \frac{mv}{qB}$.

Example:

i. Point 1:- $F = q(v+B)$
Point 2:- $F = -q(v+B)$
Point 1 force and velocity will be positive while Point 2 force will be negative.

ii.

$E = -\frac{\Delta V}{r}$

$-\Delta V = E \cdot r$

$-\Delta V = \frac{Kq}{r^2} \cdot r$

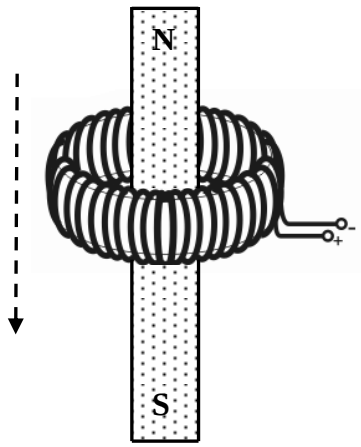
$\Delta V = Kq/r$

$\Delta V = \frac{Kq}{r}$

$r = \frac{Kq}{\Delta V}$

Question 4:

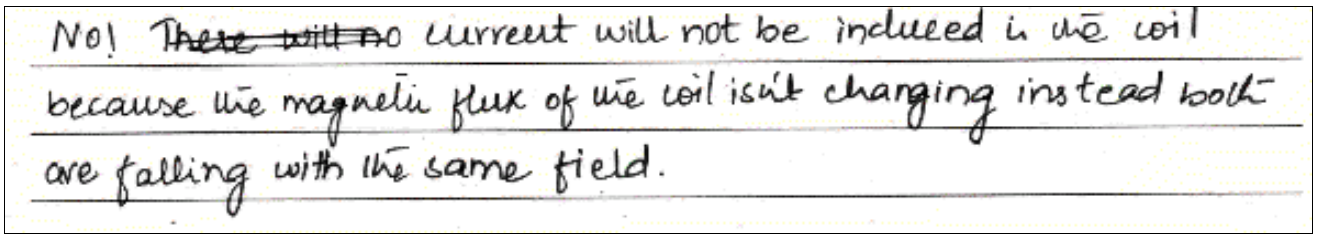
If a coil and a bar magnet are falling with the same velocity as shown in the given figure.



Will there be any induced current in the coil? Support your answer with a valid reason.

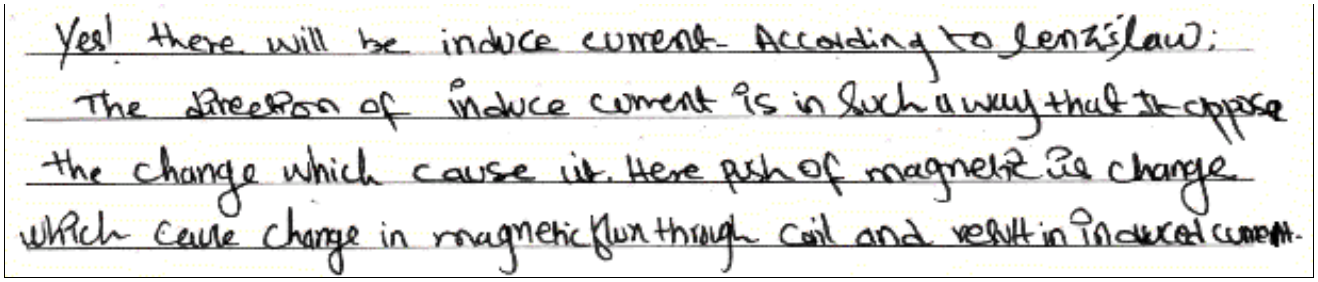
Better responses negated the possibility of any induced current in the coil. Hence, most high scoring answers mentioned 'No' and reasoned that there will be no charge of flux across the coil. Some of the high scoring candidates wrote that there will be no relative motion between the magnet and the coil, which is necessary for producing an induced current in the coil.

Example:



Weaker responses showed the weakness in candidates' understanding that both magnet and coil are falling freely and hence such candidates answered 'Yes' for the induced current and supported their answer by explaining Faraday's law of electromagnetic induction. Some of the low scoring candidates supported their explanation by writing Lenz's law which was irrelevant to the demand of the question.

Example:



Question 5:

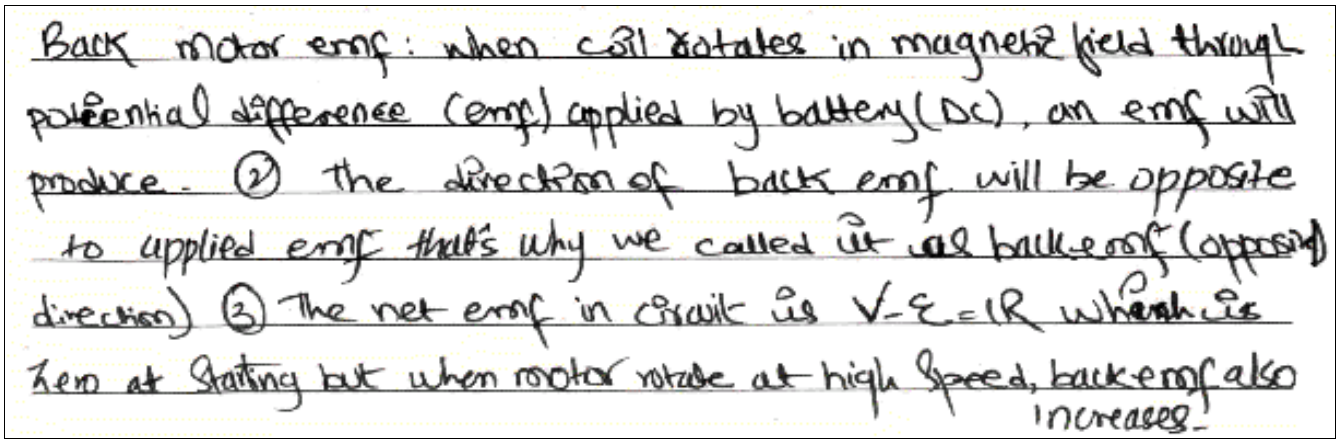
The current in an electric motor depends on both the applied electromotive force (e.m.f.) and the back electromotive force (e.m.f.).

Describe the given statement in THREE points.

Better responses described the statement by presenting knowledge of electromotive force and Lenz's law. Candidates wrote that 'the net electromotive force acts on electric motor because of opposite polarity'.

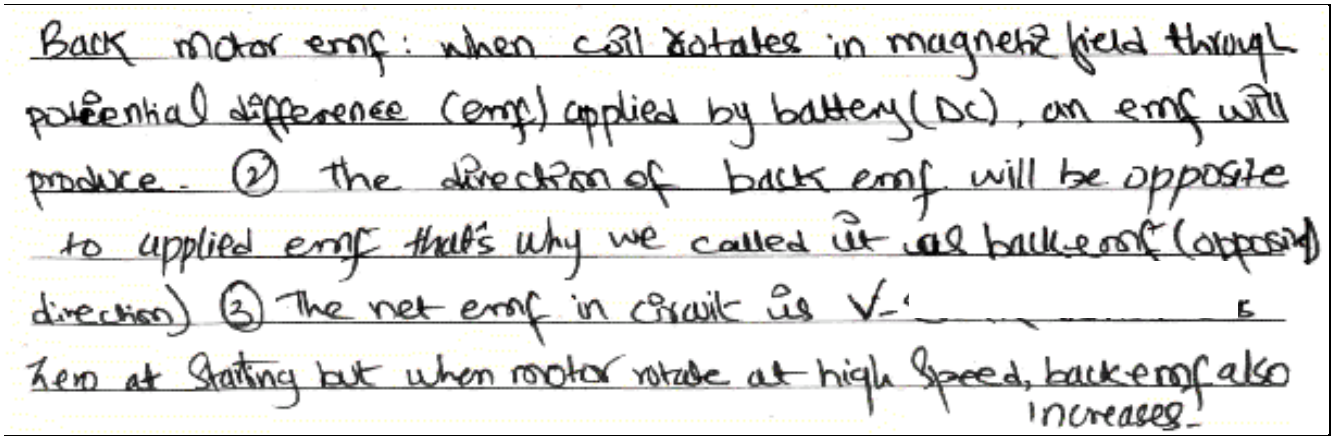
High scoring candidates wrote about the current drawn by the motor and back e.m.f. The answers indicated clarity of concept.

Example:


Back motor emf: when coil rotates in magnetic field through potential difference (emf) applied by battery (DC), an emf will produce. (2) the direction of back emf will be opposite to applied emf that's why we called it as back emf (opposite direction) (3) The net emf in circuit is $V-\mathcal{E}=IR$ which is zero at starting but when motor rotate at high speed, back emf also increases.

Weaker responses failed to describe Lenz's law and back e.m.f. clearly. Candidates wrote that the net e.m.f in the circuit is $(V-\varepsilon)$. Some of the low scoring candidates wrote about direct current, magnetic field and relative speed of the electric motor which were not demanded in the question.

Example:


Back motor emf: when coil rotates in magnetic field through potential difference (emf) applied by battery (DC), an emf will produce. (2) the direction of back emf will be opposite to applied emf that's why we called it as back emf (opposite direction) (3) The net emf in circuit is V-

zero at starting but when motor rotate at high speed, back emf also increases.

Question 6:

Describe what happens when a radiowave encounters an obstacle such as concrete barriers, rocks, mountains, etc. if the radiowave is modulated via:

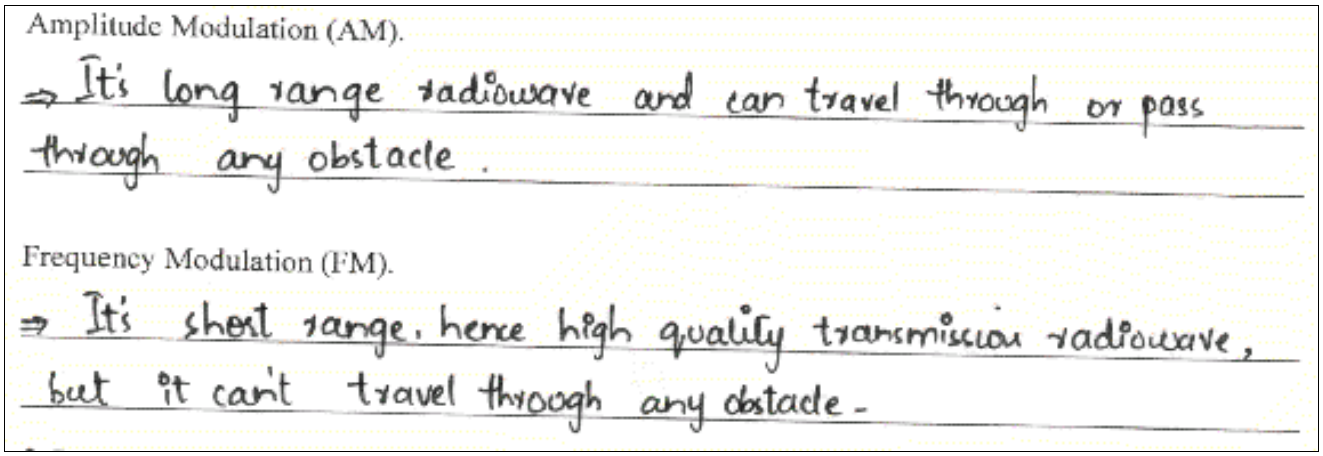
Amplitude Modulation (AM)

Frequency Modulation (FM)

This question was generally well-attempted.

Better responses correctly described the amplitude and frequency modulation. The contents of some common high scoring answers are: FM radiowaves have shorter wavelengths than to AM waves; AM radiowaves have greater wavelengths than FM waves; FM radiowaves cannot travel through obstacles like AM waves and they are highly distorted as compared to AM waves; FM radiowaves are short range waves than AM waves.

Example:


Amplitude Modulation (AM).

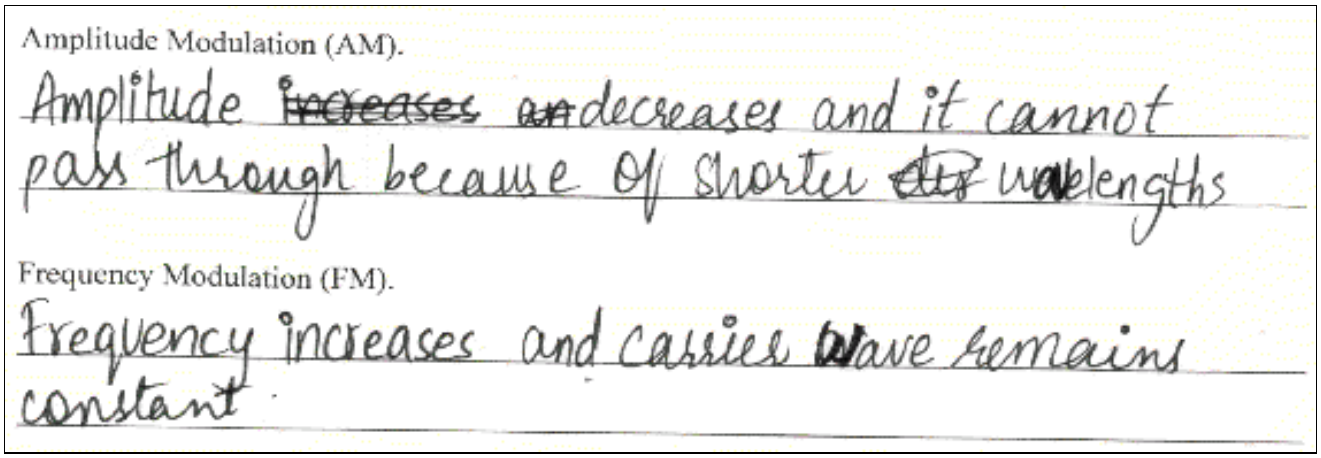
⇒ It's long range radiowave and can travel through or pass through any obstacle.

Frequency Modulation (FM).

⇒ It's short range, hence high quality transmission radiowave, but it can't travel through any obstacle.

Weaker responses failed to describe the amplitude and frequency modulation. Mostly, candidates wrote about amplitude and frequency of both radiowaves but were unable to relate them to encounter obstacles such as concrete barriers, rocks, mountains, etc.

Example:


Amplitude Modulation (AM).

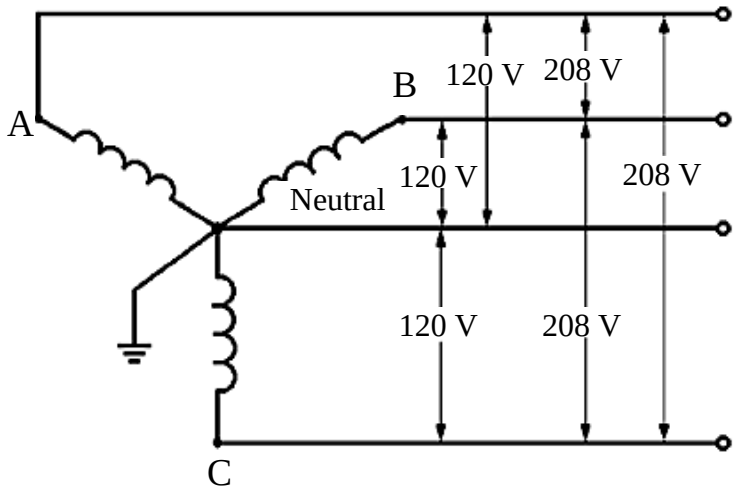
Amplitude ~~increases~~ ~~an~~decreases and it cannot pass through because of shorter ~~its~~ wavelengths

Frequency Modulation (FM).

Frequency increases and carries wave remains constant.

Question 7:

i. Identify the given circuit diagram.

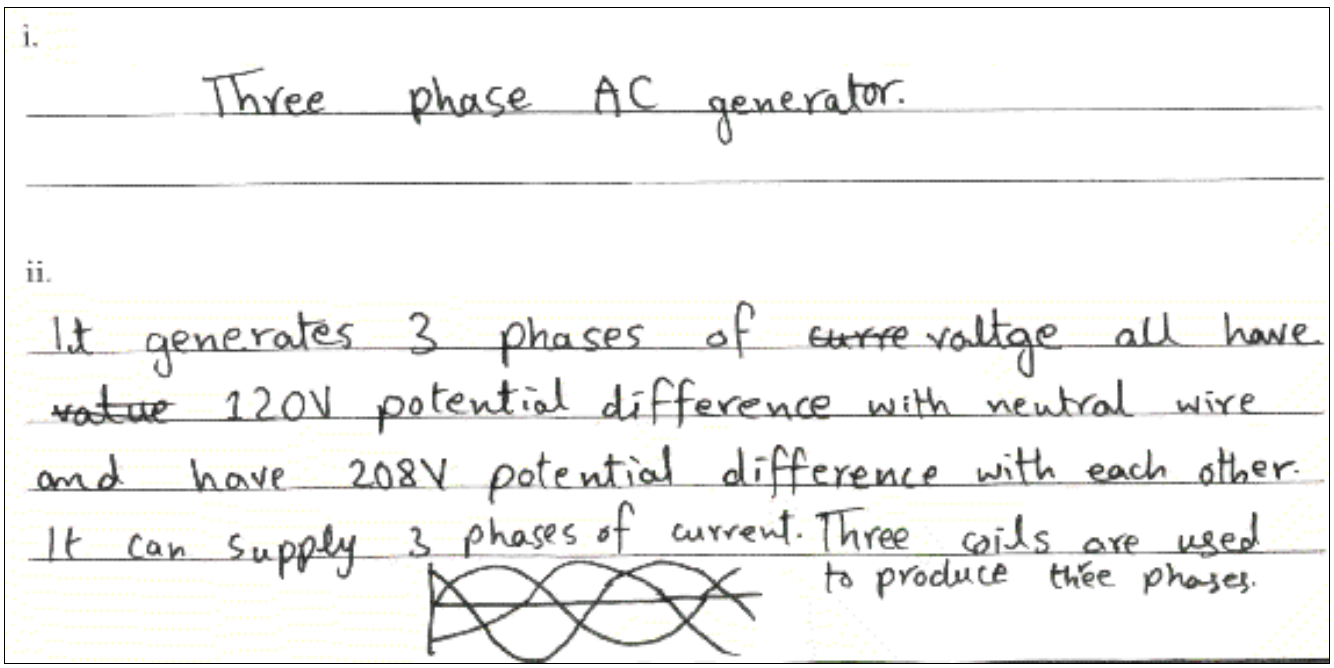


ii. Describe the working of the given circuit.

Many candidates struggled to attempt this question.

Better responses presented correct identification of the given circuit diagram as 'Three Phase A.C. Supply'. In the second part of the question, high scoring candidates correctly described the working of three phase A.C. supply, i.e. the rotation of the three coils generate three alternating voltages and the phase difference between these voltages is 120°.

Example:


i.

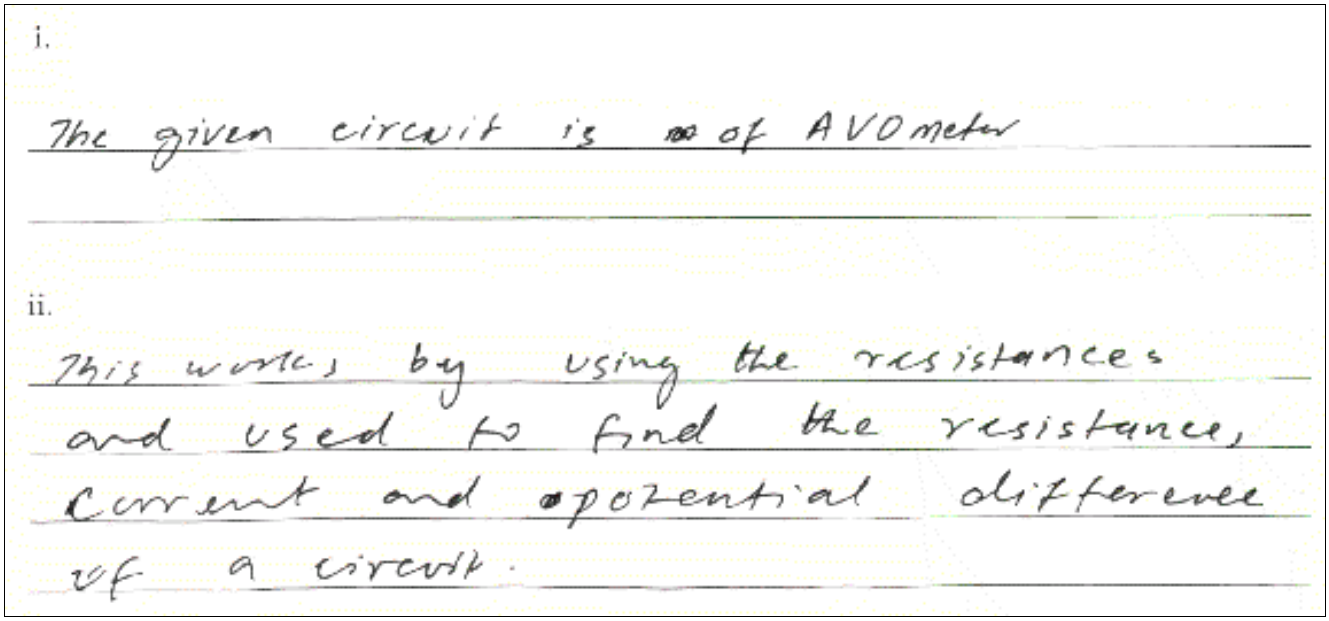
Three phase AC generator.

ii.

It generates 3 phases of ~~curre~~ voltge all have ~~value~~ 120V potential difference with neutral wire and have 208V potential difference with each other. It can supply 3 phases of current. Three coils are used to produce three phases.

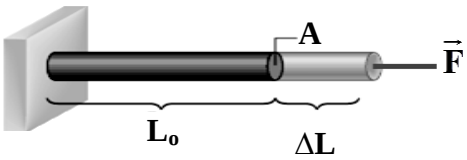
Weaker responses presented incorrect identification i.e., AVO meter or galvanometer. Some of the low scoring candidates were unable to describe the working as they wrote the relationship between current, voltage, resistance and Ohm's law which was not asked in the question.

Example:



Question 8:

In the given diagram, $\vec{\mathbf{F}}$ denotes the stretching force, **A** is the cross-sectional area, $\mathbf{L_o}$ is the original length of the rod, and $\Delta\mathbf{L}$ is the increase in length.



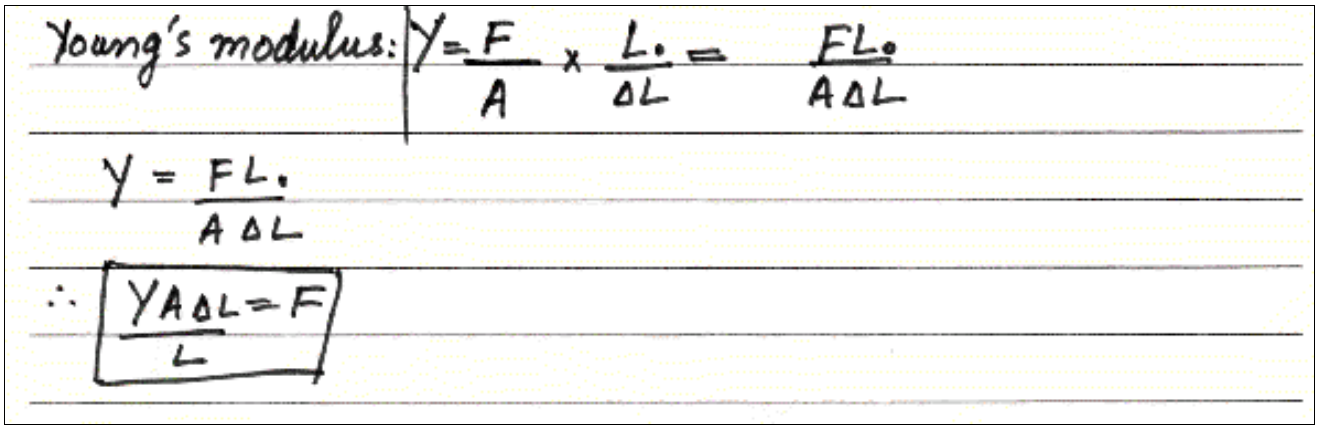
Derive the formula for the magnitude of the stretching force $\vec{\mathbf{F}}$ applied perpendicularly to the surface at the end with the help of Young's modulus.

Most candidates performed well in this question.

Better responses to the question were upto the required knowledge and majority of candidates presented their understanding of stress and strain and easily derived the formula for the magnitude of stretching force $\vec{\mathbf{F}}$ applied perpendicularly to the surface at the end with the help of Young's modulus,

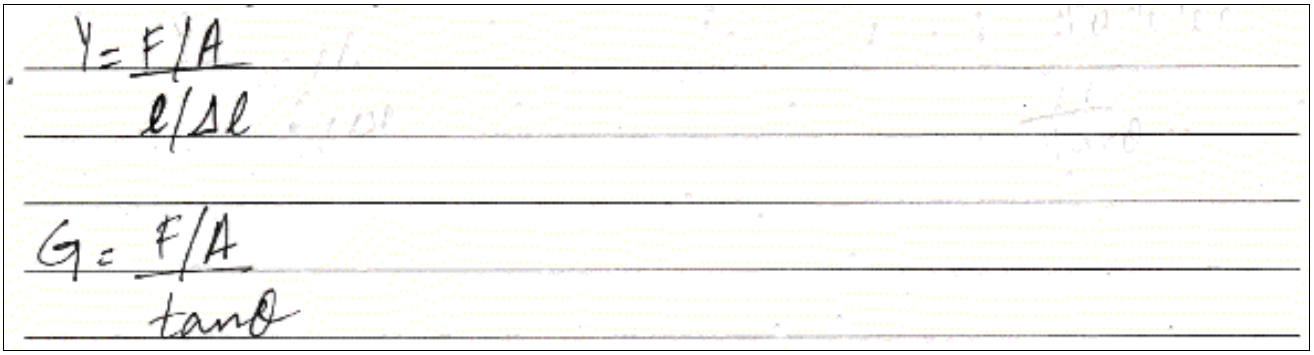
$$Y = \frac{\text{stress}}{\text{strain}}, \Rightarrow Y = \frac{\frac{F}{A}}{\frac{\Delta L}{L_0}} \text{ and } F = Y\left(\frac{\Delta L}{L_0}\right)A.$$

Example:



Weaker responses inverted the relation of stress and strain and substituted in the Young's modulus. Most low scoring candidates used tanθ, which is totally irrelevant to the derivation.

Example:



Question 9:

Fill in the blanks.

i. For AND gate, output = 1, if all inputs = _____.

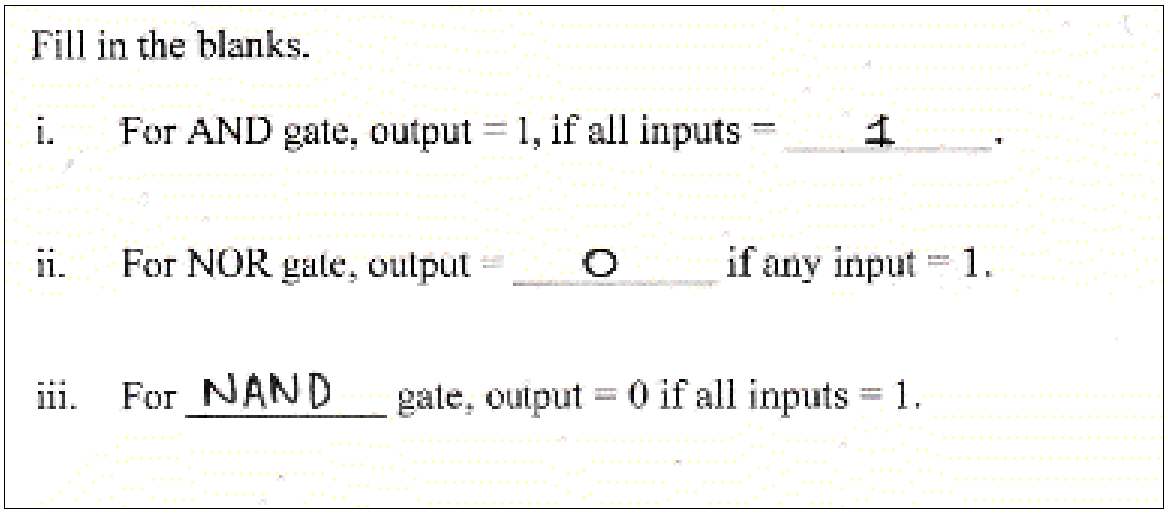
ii. For NOR gate, output = _____ if any input = 1.

iii. For _____ gate, output = 0 if all inputs = 1.

The question was generally well-attempted.

Better responses provided correct values of Boolean algebra and their function with the help of truth table with two inputs.

Example:


Fill in the blanks.

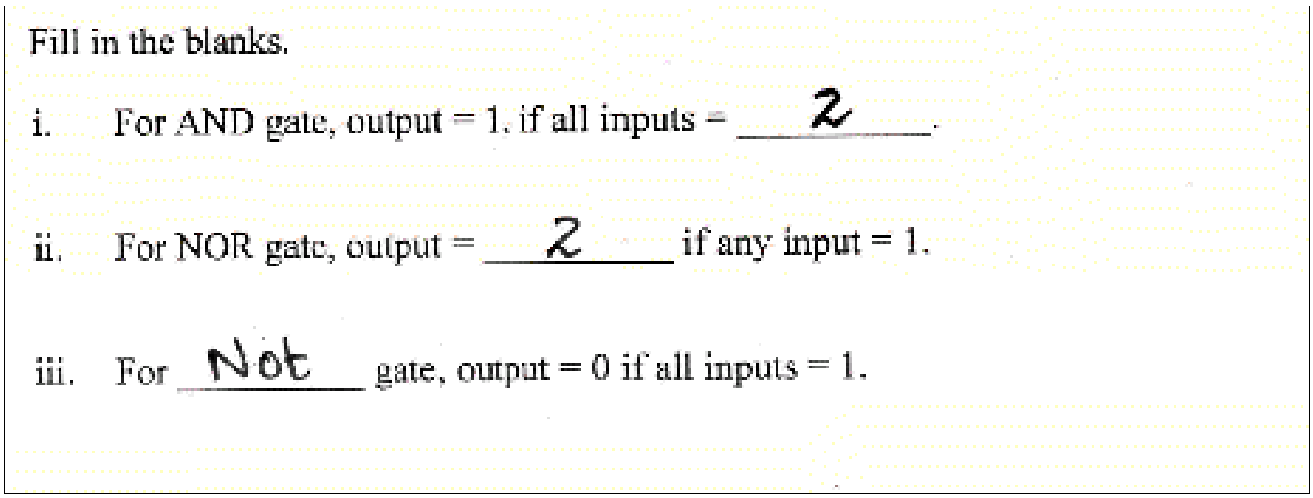
i. For AND gate, output = 1, if all inputs = 1.

ii. For NOR gate, output = 0 if any input = 1.

iii. For NAND gate, output = 0 if all inputs = 1.

Weaker responses indicated that candidates were unable to show the function of truth table with two inputs. It was observed during emarking sessions that candidates either did not touched upon this topic 'electronics' or had limited understanding of logic gates.

Example:


Fill in the blanks.

i. For AND gate, output = 1, if all inputs = 2.

ii. For NOR gate, output = 2 if any input = 1.

iii. For Not gate, output = 0 if all inputs = 1.

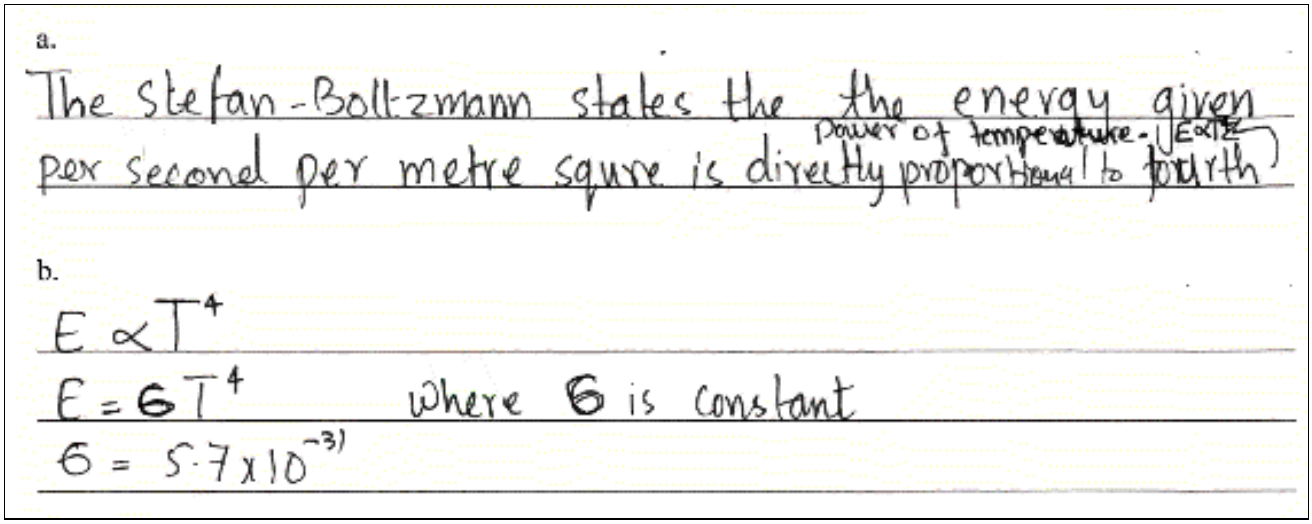
Question 10:

i. State the Stefan-Boltzmann Law of black body radiations.

ii. Write the mathematical representation of Stefan-Boltzmann Law.

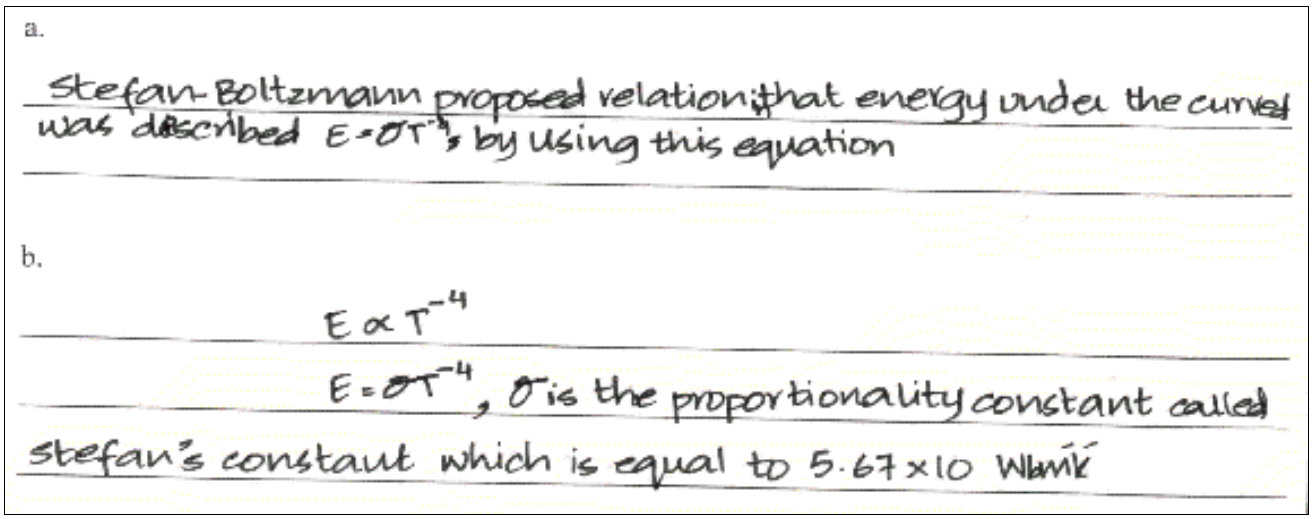
Better responses correctly stated the Stefan-Boltzmann Law of black body radiations and wrote its mathematical representation as the total energy radiated per unit surface area of a black body across all wavelengths per unit time is directly proportional to the fourth power of the black body's thermodynamic temperature T and showed in the form of mathematical expressions as, $E \alpha T^4$ and $E = kT^4$.

Example:



Weaker responses wrote the correct mathematical expressions but were unable to state the Stefan-Boltzmann Law of black body radiations correctly. Some of the low scoring candidates wrote the Einstein's theory of photoelectric effect and stated the law by using consequences of special theory of relativity, hence lost one mark for the incorrect statement.

Example:

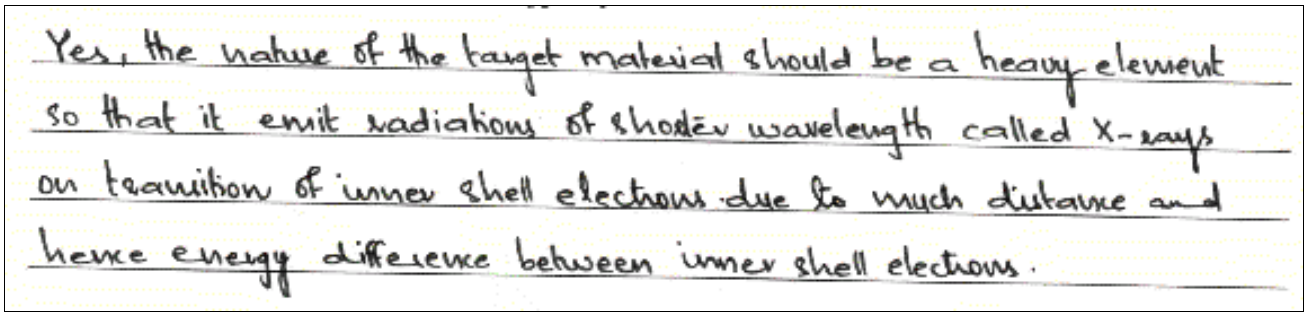


Question 11a:

Does the transition for the emission of characteristic X-rays depend on the nature of the target material atoms? Give reason to support your answer.

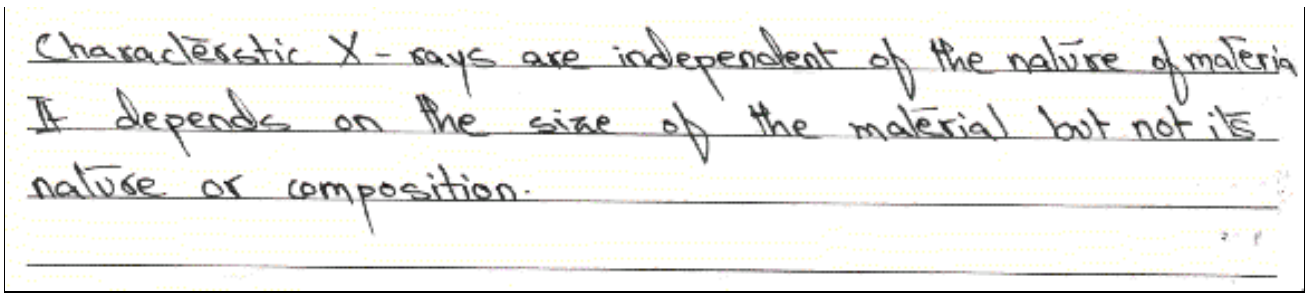
Better responses correctly wrote the reason about the concept of production of X-rays which depends upon the target surface. High scoring candidates wrote the answer 'Yes' and presented the following reason that 'the frequency, f of X-rays depends upon the atomic number (z) of the target material, $\sqrt{f} \alpha z$; materials have different number of energy levels; hence, elements have different threshold frequency'.

Example:


Yes, the nature of the target material should be a heavy element so that it emit radiations of shorter wavelength called X-rays on transition of inner shell electrons due to much distance and hence energy difference between inner shell electrons.

Weaker responses incorrectly wrote that the transition for the emission of characteristic X-rays does not depend upon surface and nature of target material atoms. Low scoring candidates wrote that characteristic of X-rays depends upon size of the material.

Example:

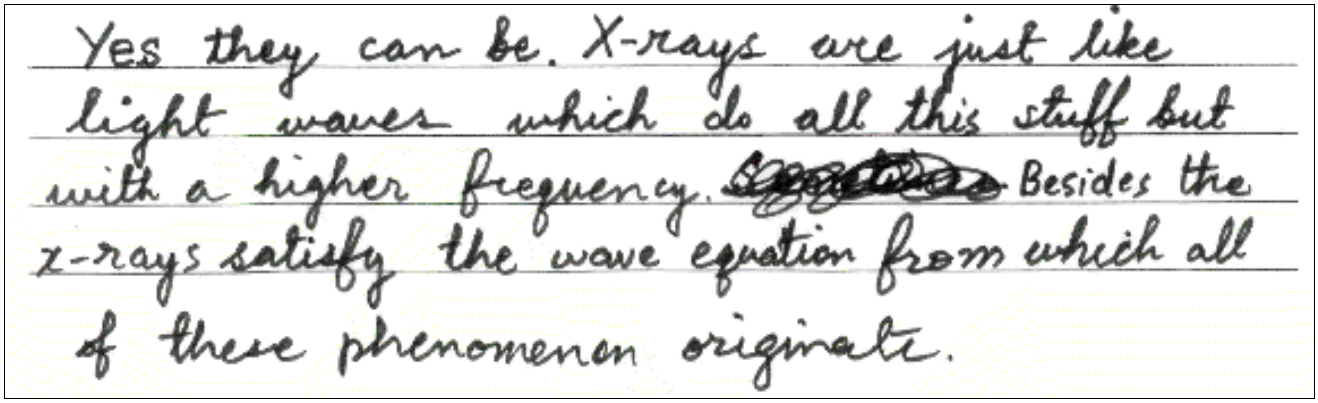

Characteristic X-rays are independent of the nature of materia
It depends on the size of the material but not its
nature or composition.

Question 11b:

Can the X-rays be reflected, refracted, diffracted and polarized just like any other waves? Give reason to support your answer.

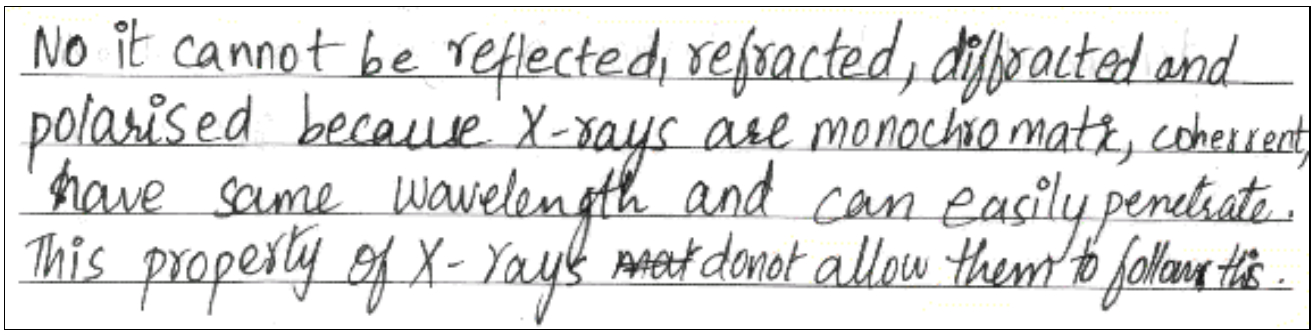
Better responses presented the correct reason that the X-rays can be reflected, refracted, diffracted and polarized just like any other waves, as X-rays are similar in nature to ordinary light with the difference of frequency and wavelength and a part of electromagnetic spectrum.

Example:


Yes they can be. X-rays are just like light waves which do all this stuff but with a higher frequency. Besides the x-rays satisfy the wave equation from which all of these phenomenon originate.

Weaker responses incorrectly wrote that X-rays cannot be reflected, refracted, diffracted and polarized just like any other waves. Low scoring candidates wrote the characteristics of X-rays that they are monochromatic, coherent and can penetrate through different materials which were not required in the question.

Example:


No it cannot be reflected, refracted, diffracted and polarised because X-rays are monochromatic, coherent, have same wavelength and can easily penetrate. This property of X-rays ~~not~~ donot allow them to follow this.

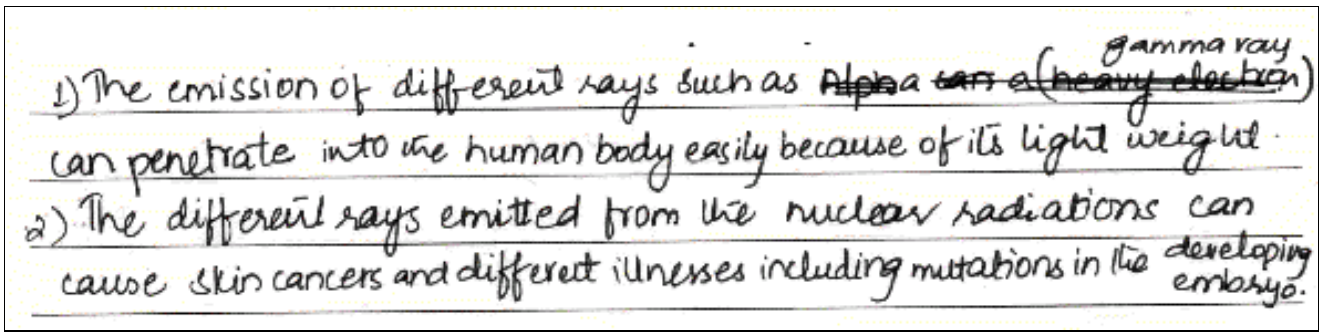
Question 12:

State TWO reasons due to which nuclear radiations are potentially harmful to humans.

Most candidates demonstrated accurate knowledge in this question.

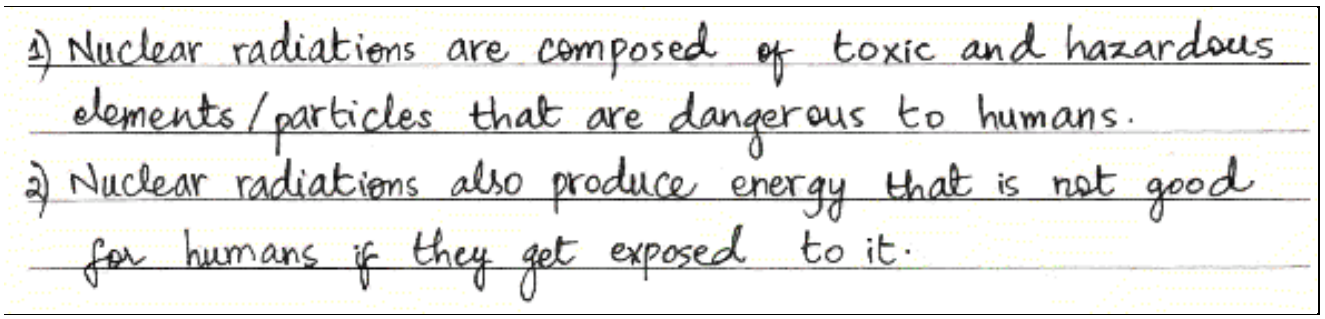
Better responses correctly stated two of the following reasons that nuclear radiations are potentially harmful to humans as, they have high ionisation power, they can alter significantly the structure of molecules within a living cell; they have high penetrating power.

Example:



Weaker responses did not try to articulate the reasons in a clear way, some candidates wrote two partially correct reasons and a few were successful in writing only one reason and hence got one mark.

Example:



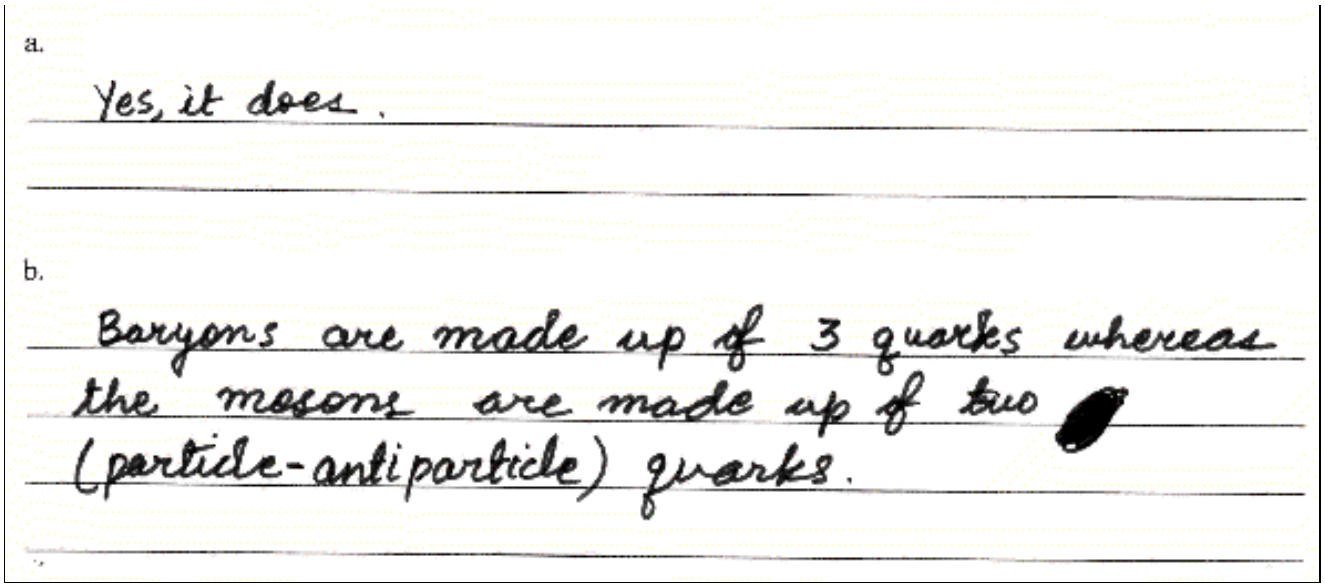
Question 13:

i. Does a quark possess a charge magnitude smaller than that of an electron?

ii. According to the quark model of hadrons, how are mesons different from baryons?

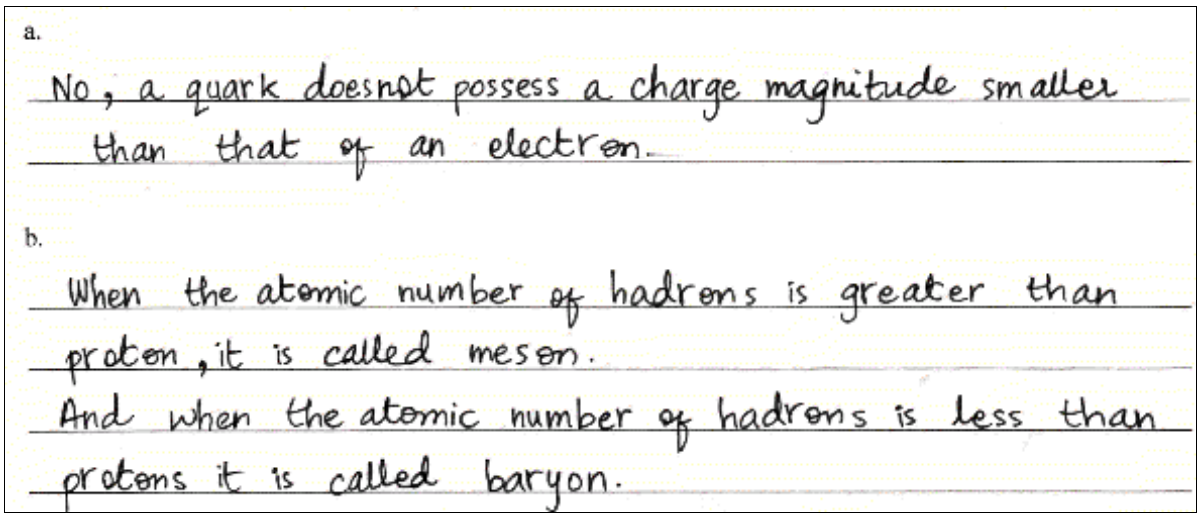
Better responses correctly answered these two parts of the questions by writing the number of quarks in mesons and baryons or with the help of weak and strong nuclear forces or referred to the lighter (mesons) and heavier (baryons) particles.

Example:



Weaker responses showed candidates' weak knowledge of fractional charge which possesses by quarks. Some of the low scoring candidates were also unable to provide the differences between mesons and baryons.

Example:



Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**

(**Note**: Majority of the candidates attempted Question **14b** and they scored well in this part.)

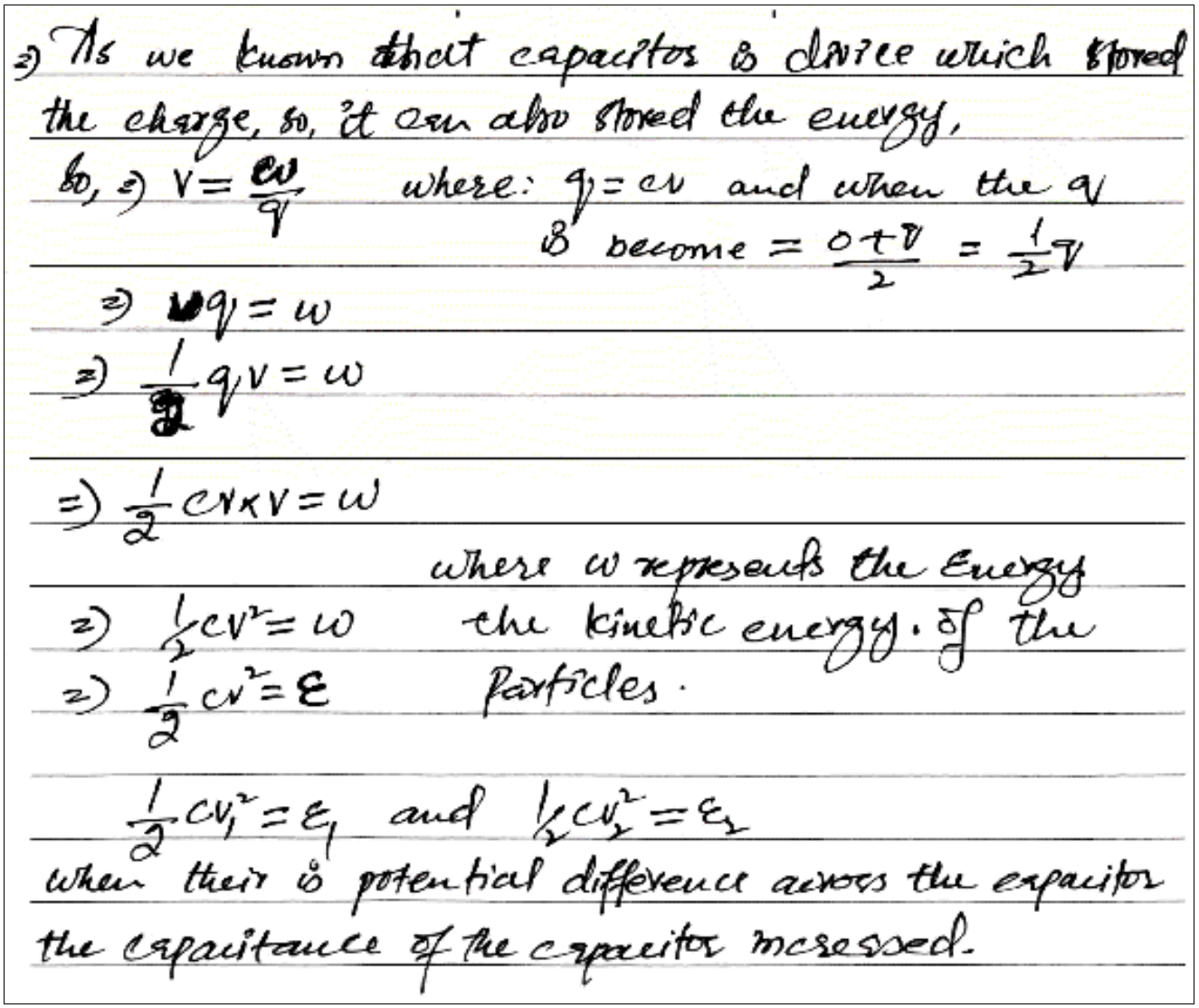
Question 14a:

i. Discuss the effect observed after the application of potential difference across the plates of a capacitor.

ii. Derive the expression for the energy **E** stored in an uncharged capacitor taking voltage at first plate as $\mathbf{V_1}$ and voltage at second plate as $\mathbf{V_2}$.

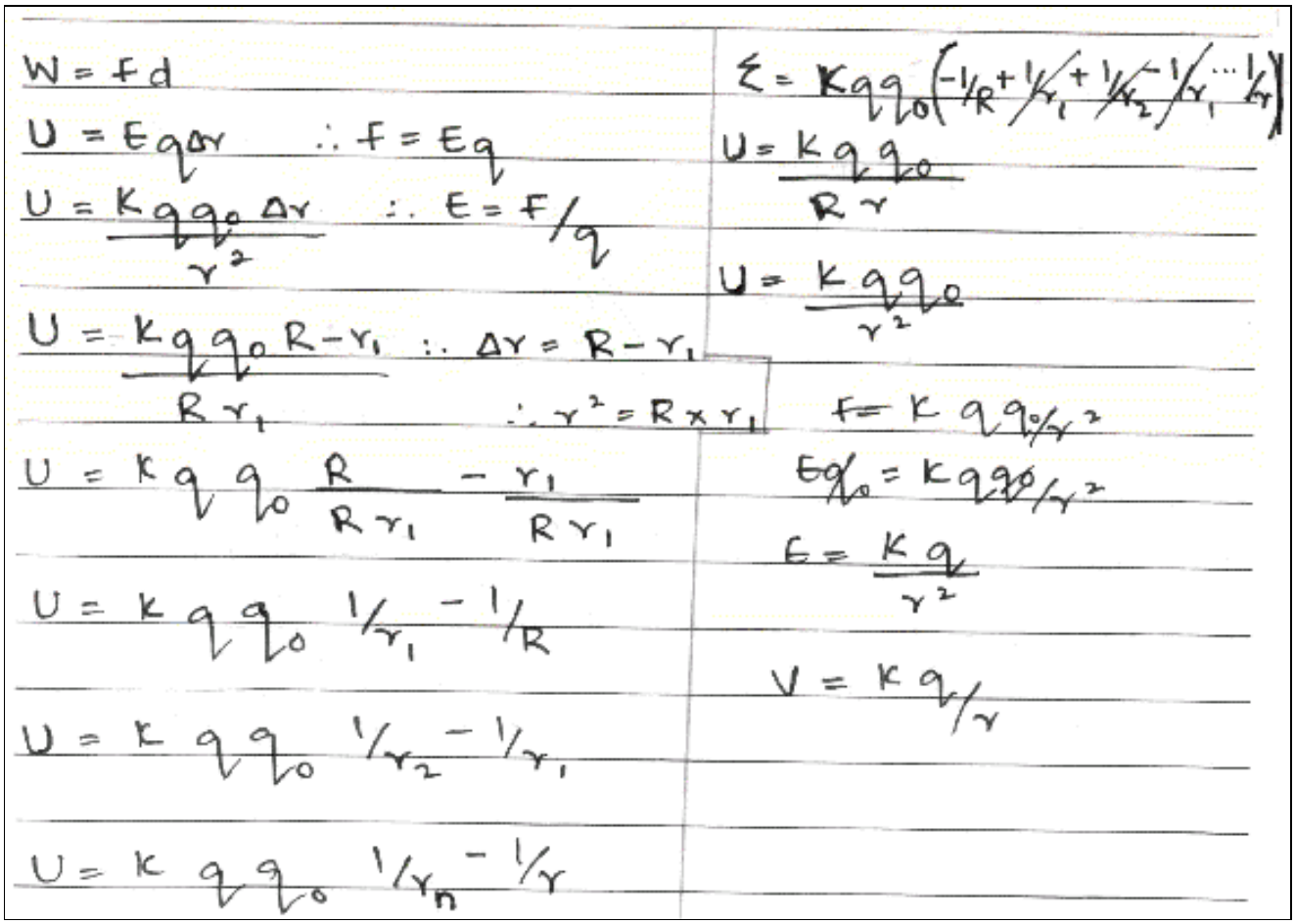
Better responses correctly discussed the concept of polarisation between two plates of a parallel plate capacitor and also derived the expression for the energy stored in a capacitor by writing the mathematical expressions, $V_{avg} = \frac{1}{2}V$, $Q = C\,V$ and $W = \frac{1}{2}CV^2$.

Example:



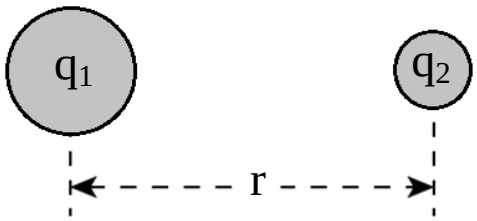
Weaker responses failed to discuss the concept of polarisation and derived the expression for the energy stored in a capacitor. Low scoring candidates erroneously discussed the polarisation between two plates of a parallel plates capacitor; they wrote the mathematical relation of Coulomb's law and hydrogen spectrum in terms of different energy levels which were totally irrelevant to the requirement of the question.

Example:



Question 14b:

State and explain the law which describes the relation of force between two charged particles q_1 and q_2 and write the value of electrostatic constant.



Better responses correctly stated and explained the Coulomb's law by mentioning the mathematical expressions as $F \propto \frac{q_1 q_2}{r^2}$, $F = k\frac{q_1 q_2}{r^2}$ and the constant of proportionality, $\left(k = \frac{1}{4\pi\pi_o} = 9\times10^9\ \mathrm{Nm^2C^{-2}}\right)$.

Example:

Coulumnb's law:-

The force (attraction or repulsion) between two point charges is directly proportional to the product of the magnitud of charges and inversely Proportional to the sqware of the distance between them.

$F \propto q_1 q_2$ — (1)

$F \propto \frac{1}{r^2}$ — (2)

combining (1) and (2) $F \propto \frac{q_1 q_2}{r^2}$

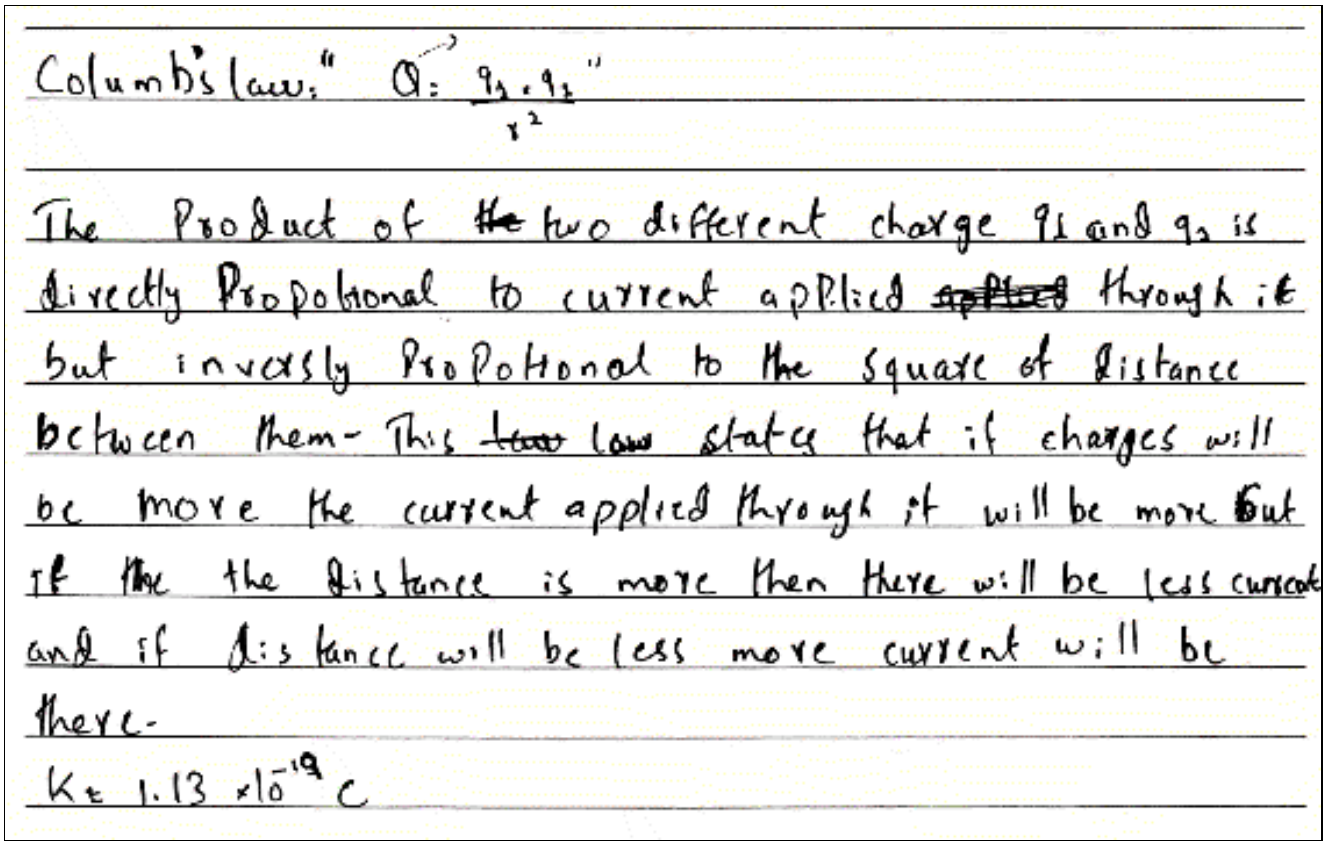
$F = k\frac{q_1 q_2}{r^2}$

k is the constant of Proportionality is called electrostatic constant.

$k = \frac{1}{4\pi\varepsilon_0} = 9\times10^9 \times \mathrm{Nm^2C^{-2}}$

Weaker responses failed to state and explain the Coulomb's law, whereas some of the low scoring candidates either wrote correct statement or the correct equation.

Example:


Columb's law: "$Q = \frac{q_1 \cdot q_2}{r^2}$"

The Product of ~~the~~ two different charge q_1 and q_2 is directly Proportional to current applied ~~applied~~ through it but inversely Proportional to the square of distance between them- This ~~law~~ law states that if charges will be more the current applied through it will be more but if the the distance is more then there will be less current and if distance will be less more current will be there-

K = 1.13×10^{-19} C

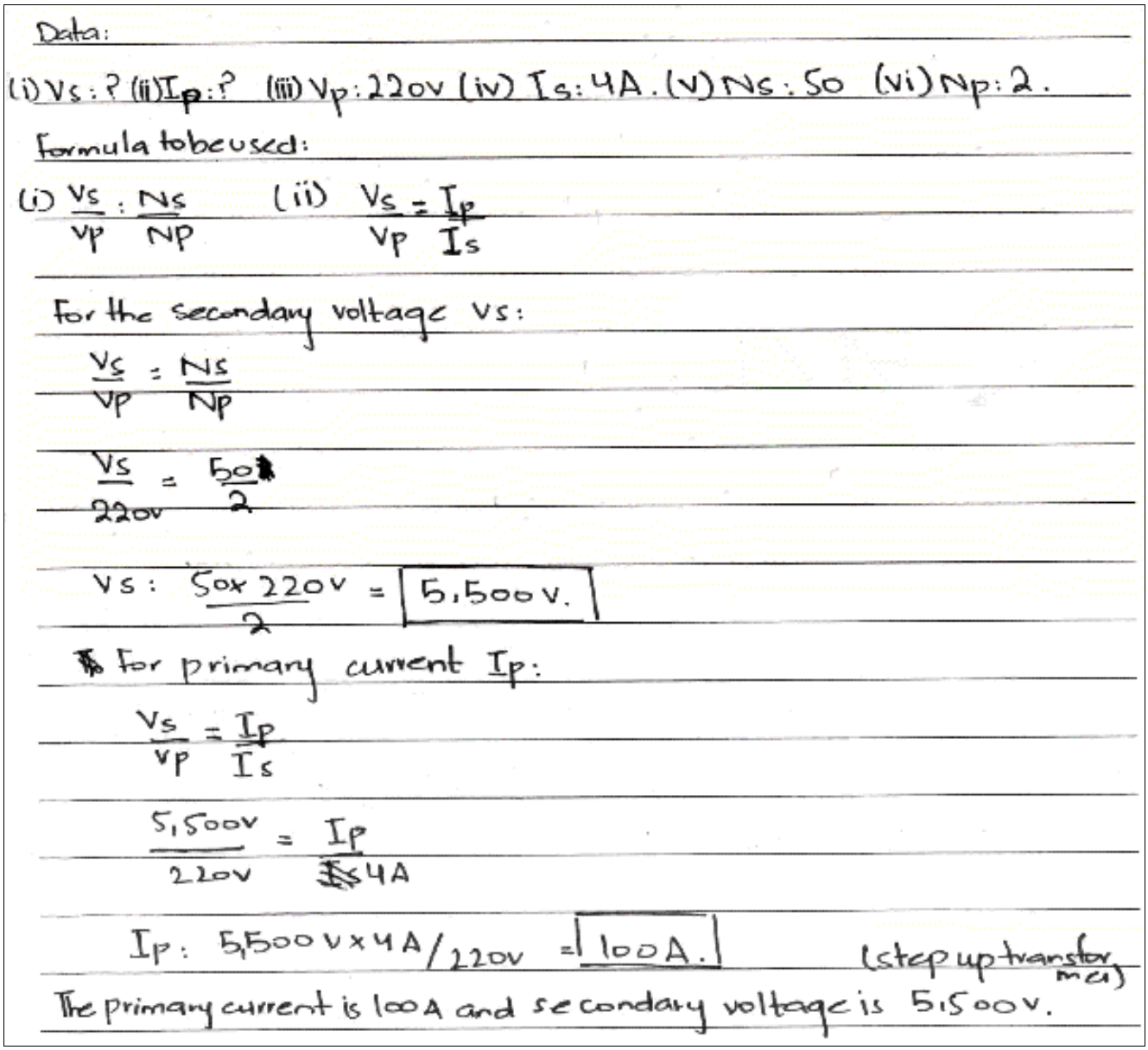
(**Note**: Majority of the candidates attempted Question **15a** and they scored well in this part.)

Question 15a:

A transformer works on 220 V and supplies a load with 4 A. If the ratio of the primary and secondary winding of coils is 2:50 respectively, then calculate the (i) primary current and (ii) secondary voltage in the coils.

Better responses correctly extracted data (values) from the question, used the correct formula for calculating primary current, $\frac{N_s}{N_p} = \frac{I_p}{I_s}$ and hence obtained the result, I_p = 100 A. In the second part of the question, high scoring candidates calculated the secondary voltage by the formula, $\frac{V_s}{V_p} = \frac{I_p}{I_s}$ and finally got the correct answer, V =5500 V.

Example:



Weaker responses correctly used either both formulae for calculating primary current and secondary voltage in the coils or used only one formula for secondary voltage, some of the low scoring candidates were unable to correctly substitute the values in the formulae, they used the value of current in place of voltage and got the incorrect answer of primary current I_p, likewise few of the candidates failed to reach the correct answer for secondary voltage, V_s.

Example:

As

$\frac{Ns}{Np} = \frac{Is}{Ip}$

$\frac{50}{2} = \frac{4}{Ip}$

$25 = \frac{4}{Ip}$

$\frac{25}{4} = \frac{1}{Ip}$

$\frac{4}{25} = Ip$

$0.16\ A = Ip$

Put the value of Ip in equ(B)

$\frac{Vs}{Vp} = \frac{Ip}{Is}$ — equ(B)

$Vs = \frac{Ip \times Vp}{Is}$

$Vs = \frac{0.16}{4} \times 220$

$Vs = 8.8V$

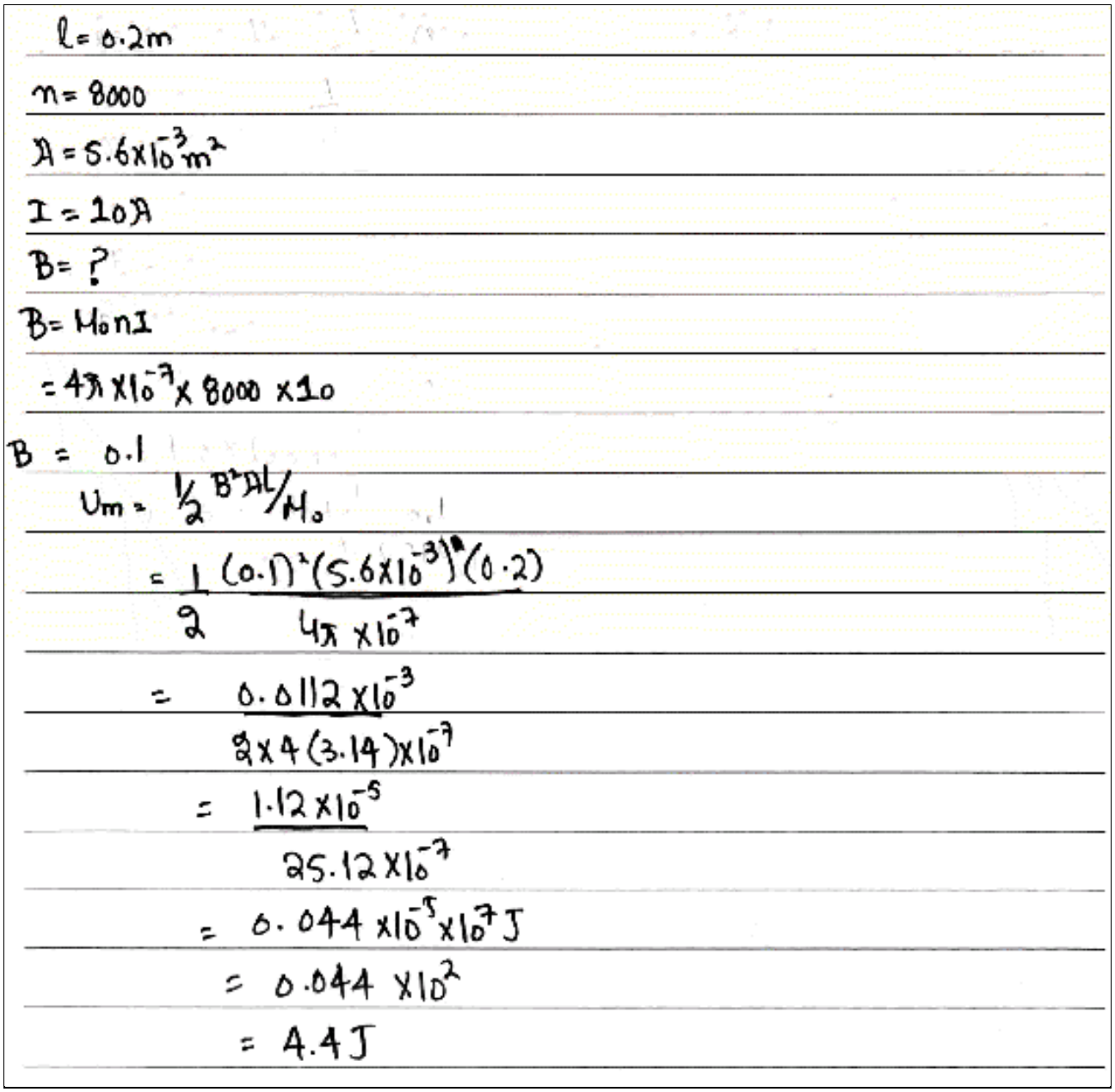
Question 15b:

A 0.2 m long inductor has 8000 turns per metre and its cross-sectional area is 5.6 $\times 10^{-3}$ m^2. If 10 A current is passing through the inductor, then calculate the value of energy stored in it.

(**Note**: The value of $\mu_o = 4\pi \times 10^{-7}$ Wb/A.m)

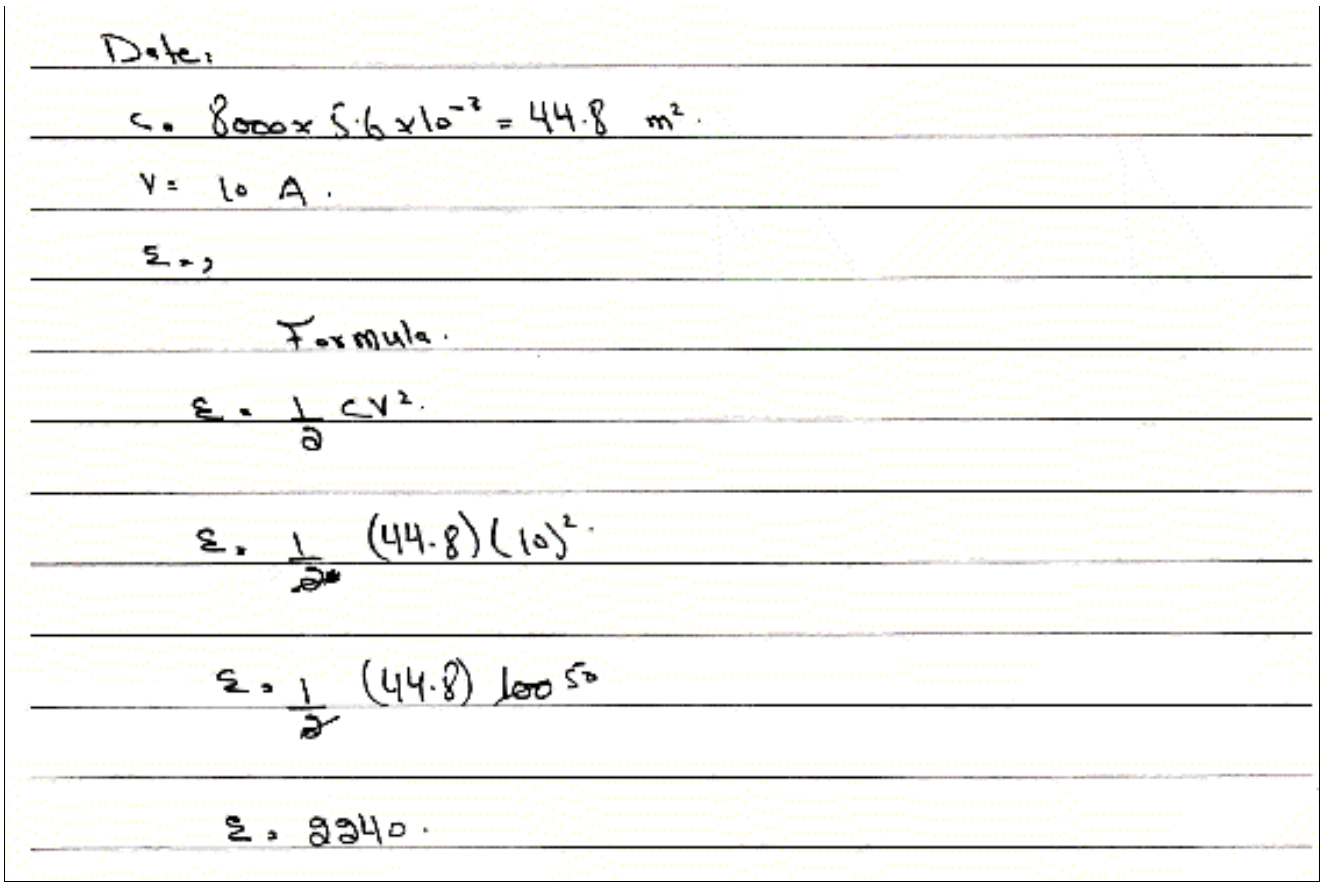
Better responses correctly calculated the value of energy stored in an inductor by using the formula, $L = \mu_o n^2 A l$, and obtained the value of L, 9.0 $\times 10^{-2}$ Wb/A, and finally used the formula for energy stored in an inductor, $E = \frac{1}{2} L I^2$, to calculate the correct value of E, 4.5 J.

Example:



Weaker responses failed to write the correct formula for the energy stored in an inductor; candidates appeared confused and used the formula for the energy stored in a capacitor, $W = \frac{1}{2}CV^2$. Most of the low scoring candidates only extracted the correct data (values) from the question without proceeding any further.

Example:



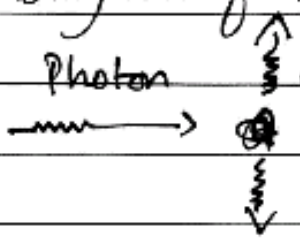
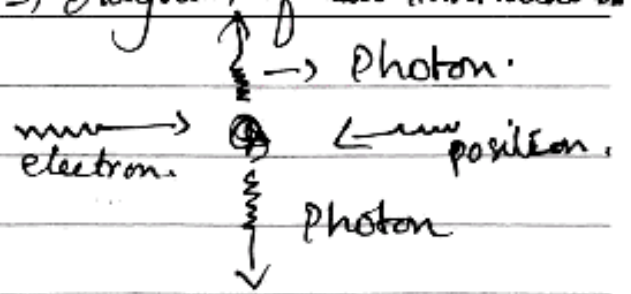
(**Note**: Majority of the candidates attempted Question **16a** and they scored well in this part.)

Question 16a:

Compare mechanisms of pair production and pair annihilation in FIVE points.

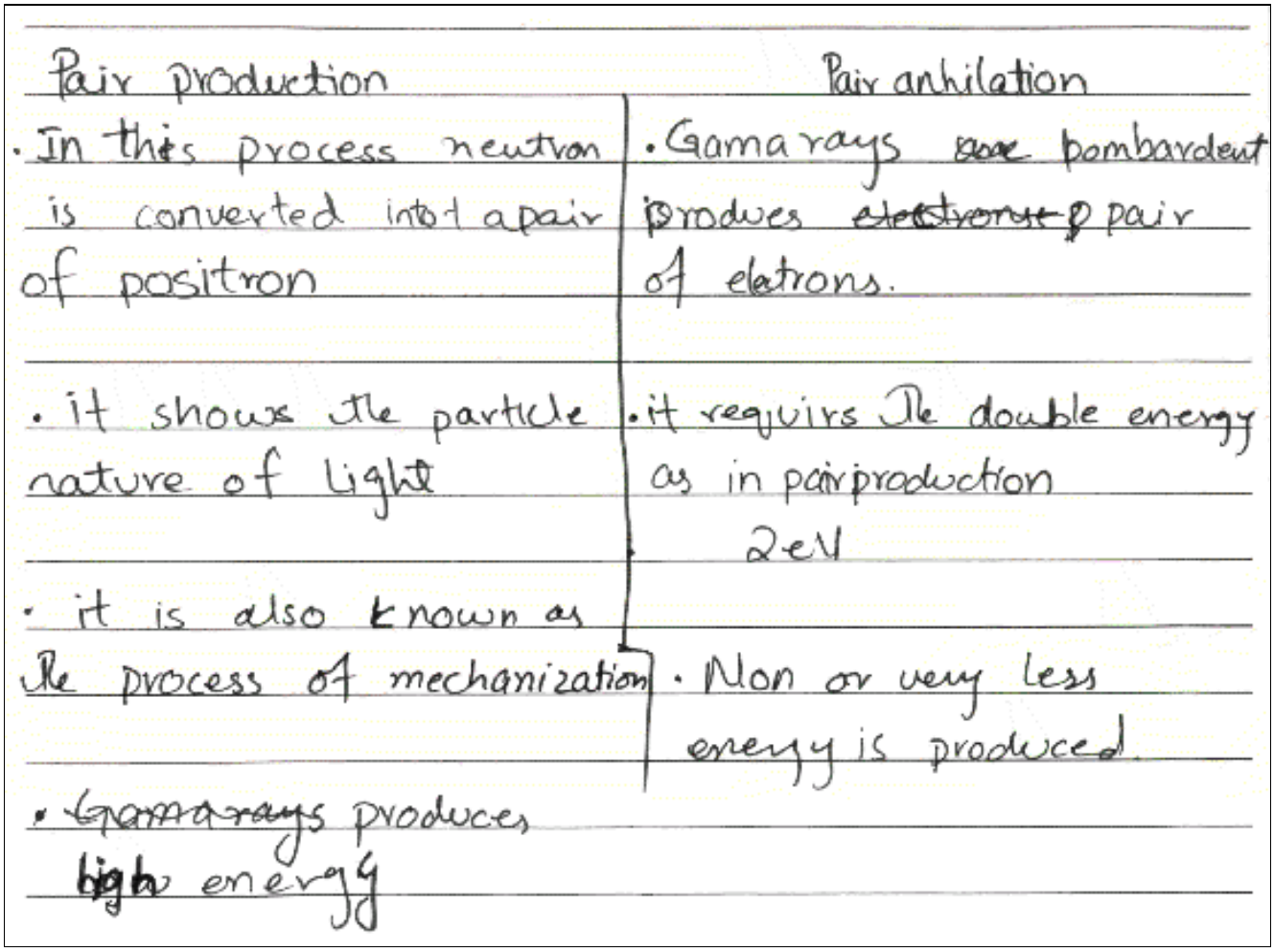
Better responses correctly compared the mechanisms of pair production and pair annihilation; candidates compared both the phenomenon as pair production takes place when a very high energy photon such as that of γ-rays interacts with matter, whereas a particle and antiparticle come together, they annihilate each other and only energy is left; in pair production photon energy is changed into an electron-position pair, whereas in pair annihilation, energy can also be used to create other particles; a position is a particle having mass and charge equal to that of an electron but charge being opposite to that of electron, whereas particles can be created by converting the energy into photon to a particle-antiparticle.

Example:

Pair Production	Pair Annihilation
=> Under certain circumstance when photon produces Positron and electron is called pair production.	Under certain circumstance when electron and positrons disappears to produce two photon called Pair Annihilation.
=> This Pair Production is materilixation	=> Pair Annihilation is de-materilixation.
=> In Pair production only one photon produces electron and positron.	=> In Pair Annihilation electron and positron produces two photons.
=> Diagram of Process of Pair – P. 	=> Diagram of Pair Annihilation 
=> In Pair Production no interference took place.	=> In Pair Annihilation interference took Place.

Weaker responses wrote about nuclear radiations, fusion and fission reaction and energy carrying particles which were not demanded in the question.

Example:



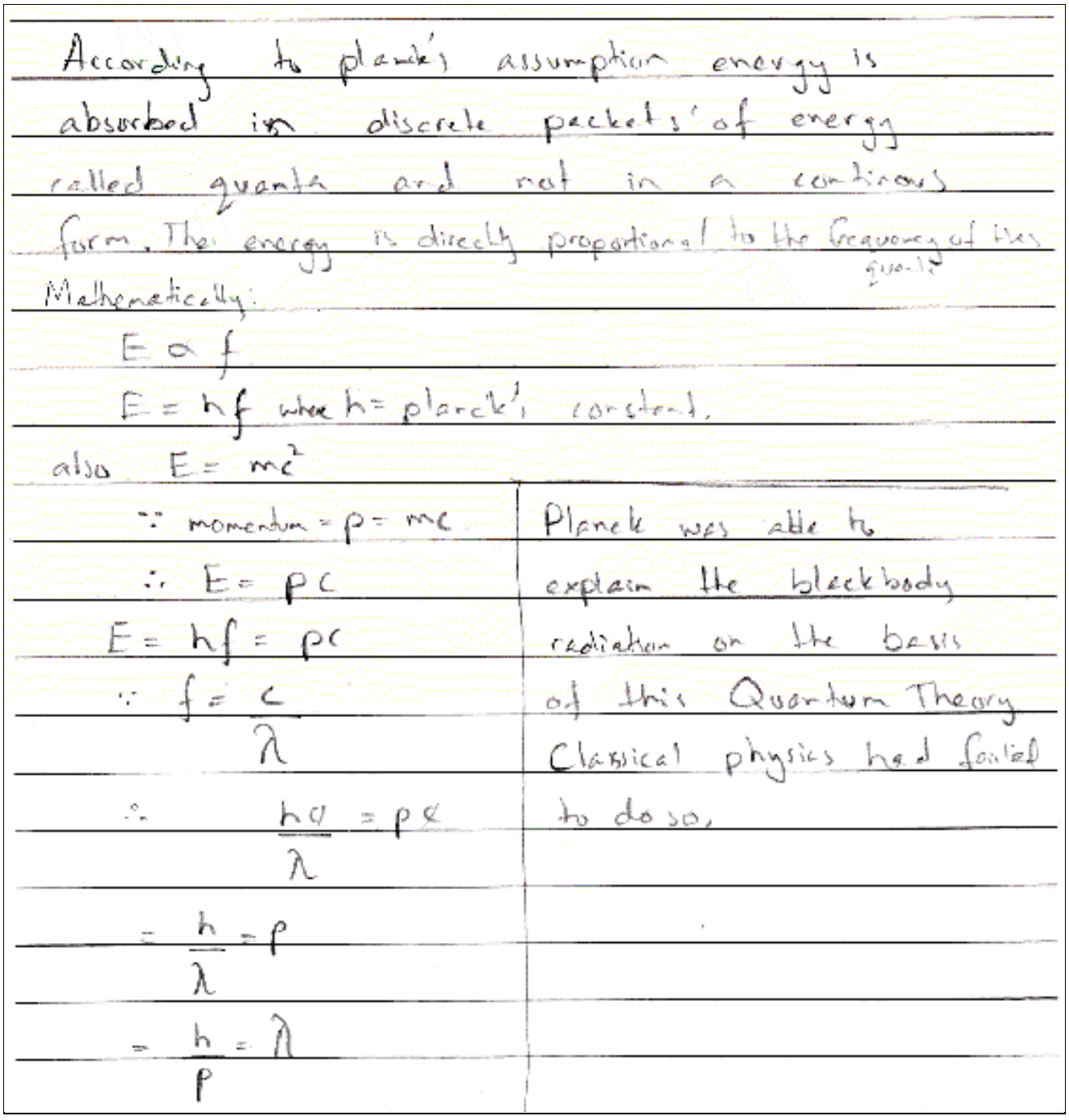
Pair production	Pair anhilation
• In this process neutron is converted into a pair of positron	• Gama rays ~~are~~ bombarded produces ~~electrons~~ pair of elatrons.
• it shows the particle nature of light	• it requires the double energy as in pairproduction 2eV
• it is also known as the process of mechanization	• Non or very less energy is produced.
• ~~Gamarays~~ produces ~~high~~ energy	

Question 16b:

Describe Planck's assumption about the black body radiation with the help of mathematical relation on energy and momentum.

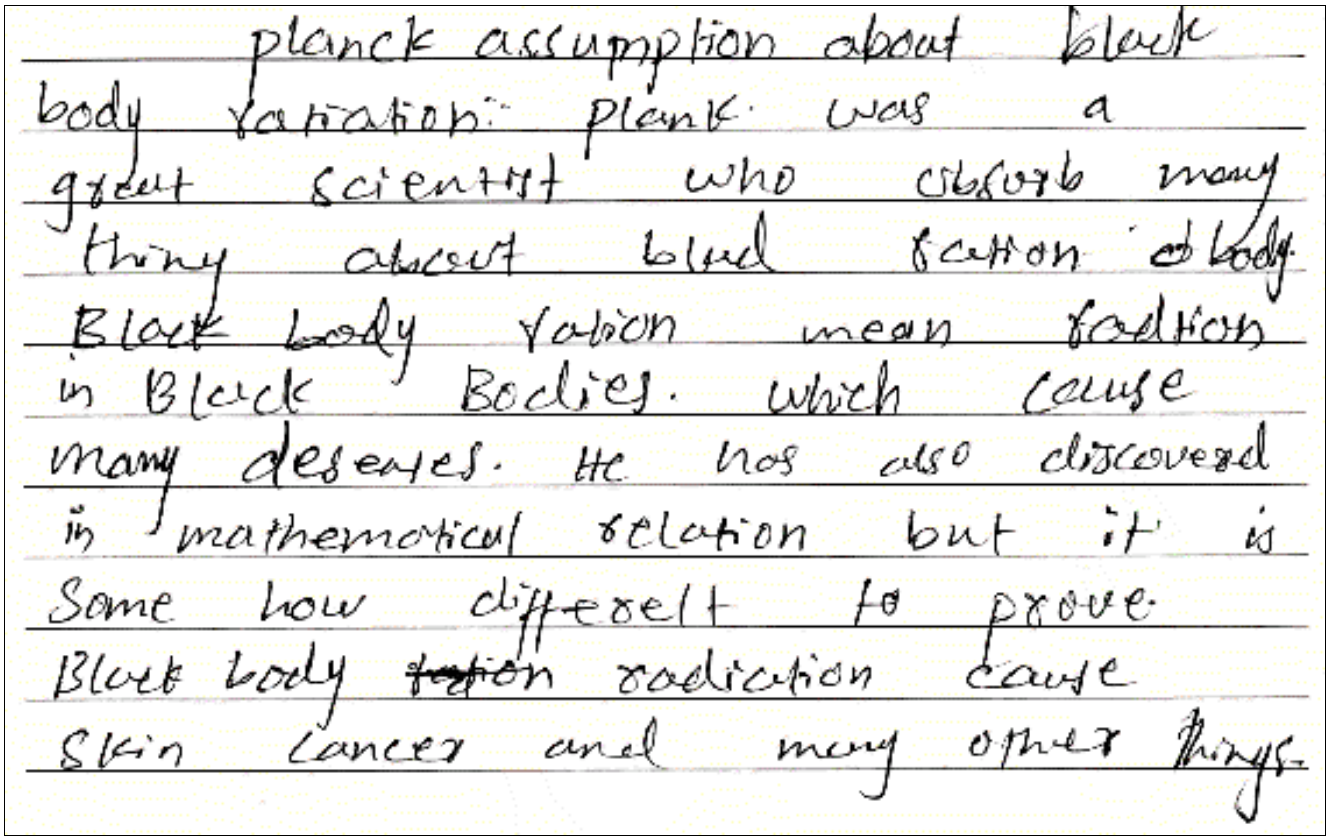
Better responses described Planck's assumption about the black body radiation with the help of mathematical relation on energy and momentum, high scoring candidates described that the energy is radiated or absorbed in the form of discrete particles called quanta Q, they also described the momentum of these particles by the mathematical relations, $E \alpha f$, $E = hf$, $E = pc$ and $p = \frac{hf}{c} = \frac{h}{\lambda}$.

Example:



Weaker responses described incorrect and irrelevant Planck's assumption about the black body radiation. Low scoring candidates wrote about diseases caused by nuclear radiations and how harmful these radiations are for living beings, these points are totally irrelevant to the demand of the question.

Example:


planck assumption about black body radiation: Plank was a great scientist who absorb many thing about blue ration of body. Black body ration mean radiation in Black Bodies. which cause many deseases. He has also discovered in mathematical relation but it is some how difficult to prove. Black body ~~ration~~ radiation cause skin cancer and many other things.